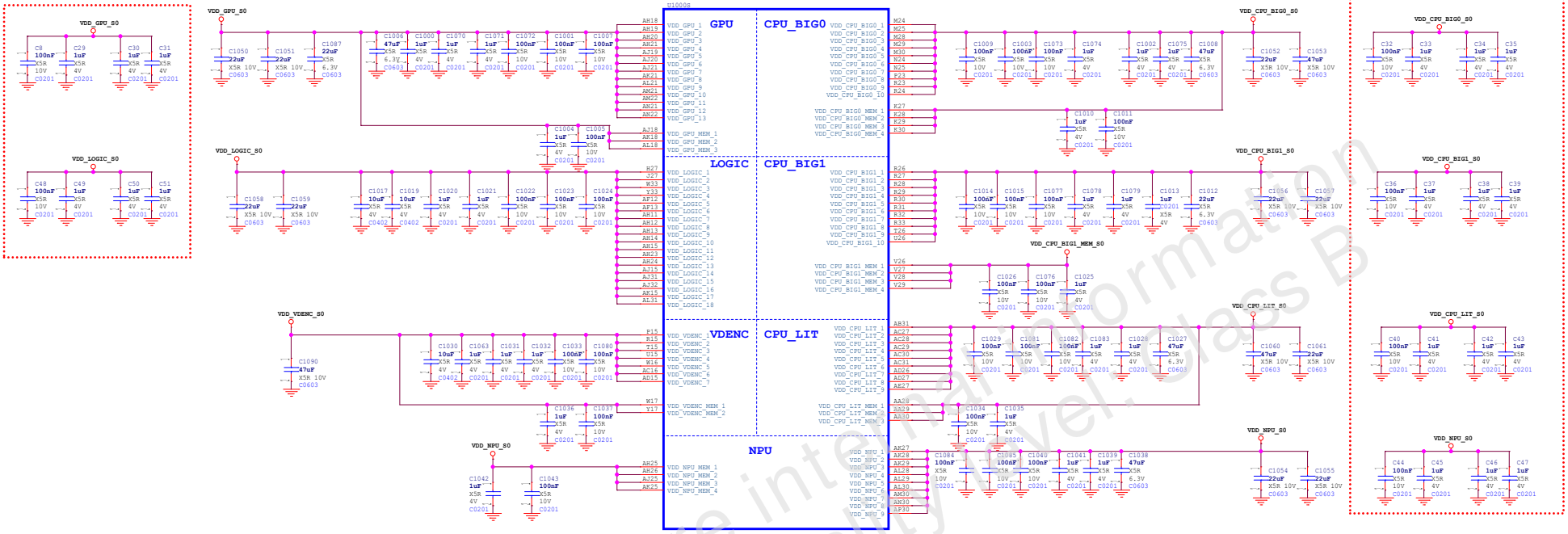
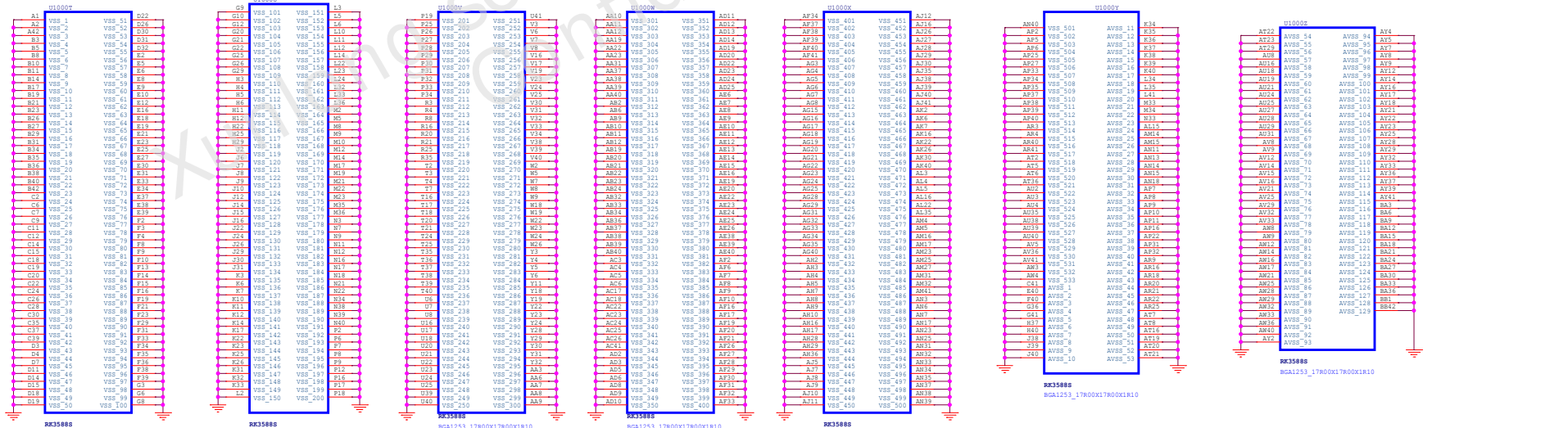
RK3588S(Power&Gnd)





Xunlong software internal information
Confidentiality level: Class B

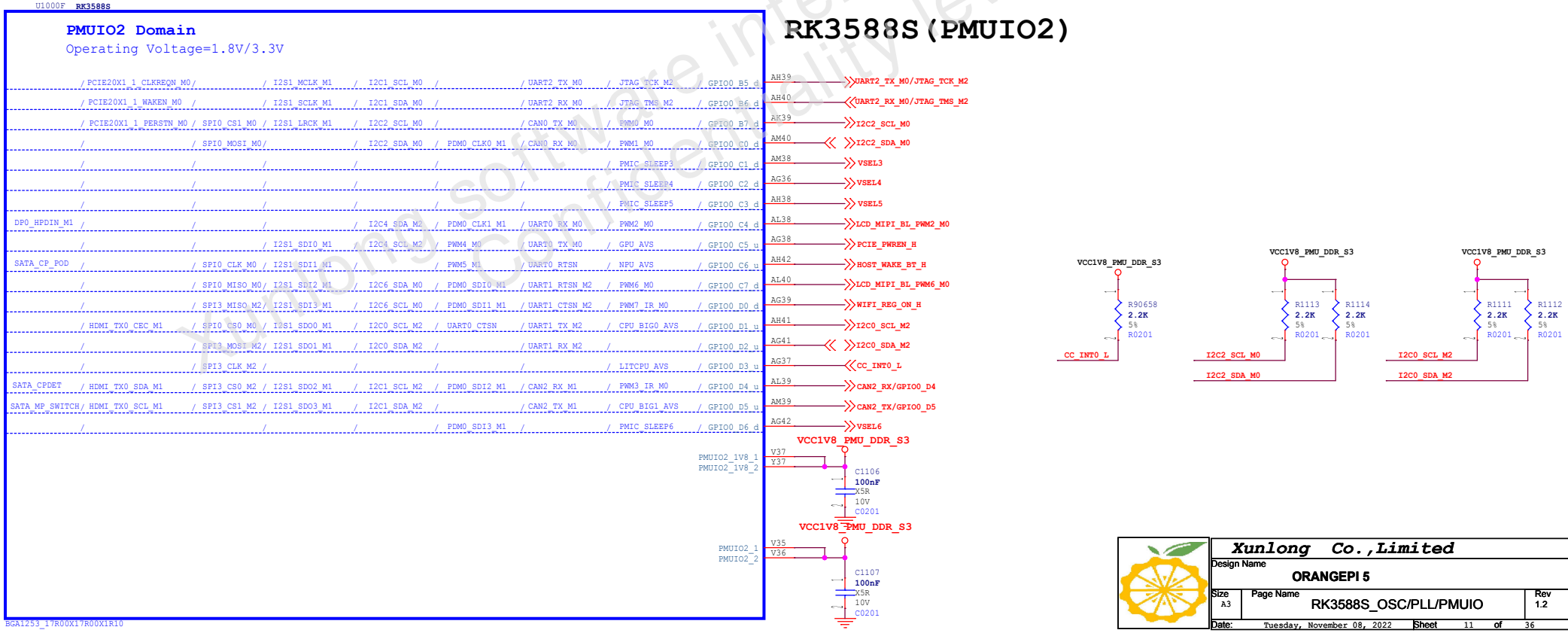
Xunlong Co.,Limited

Design Name	ORANGEPI 5	
Size: A2	Page Name: RK3588S_Power/GND	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 10 of 36	

RK3588S(OSC/PLL/PMUIO1)



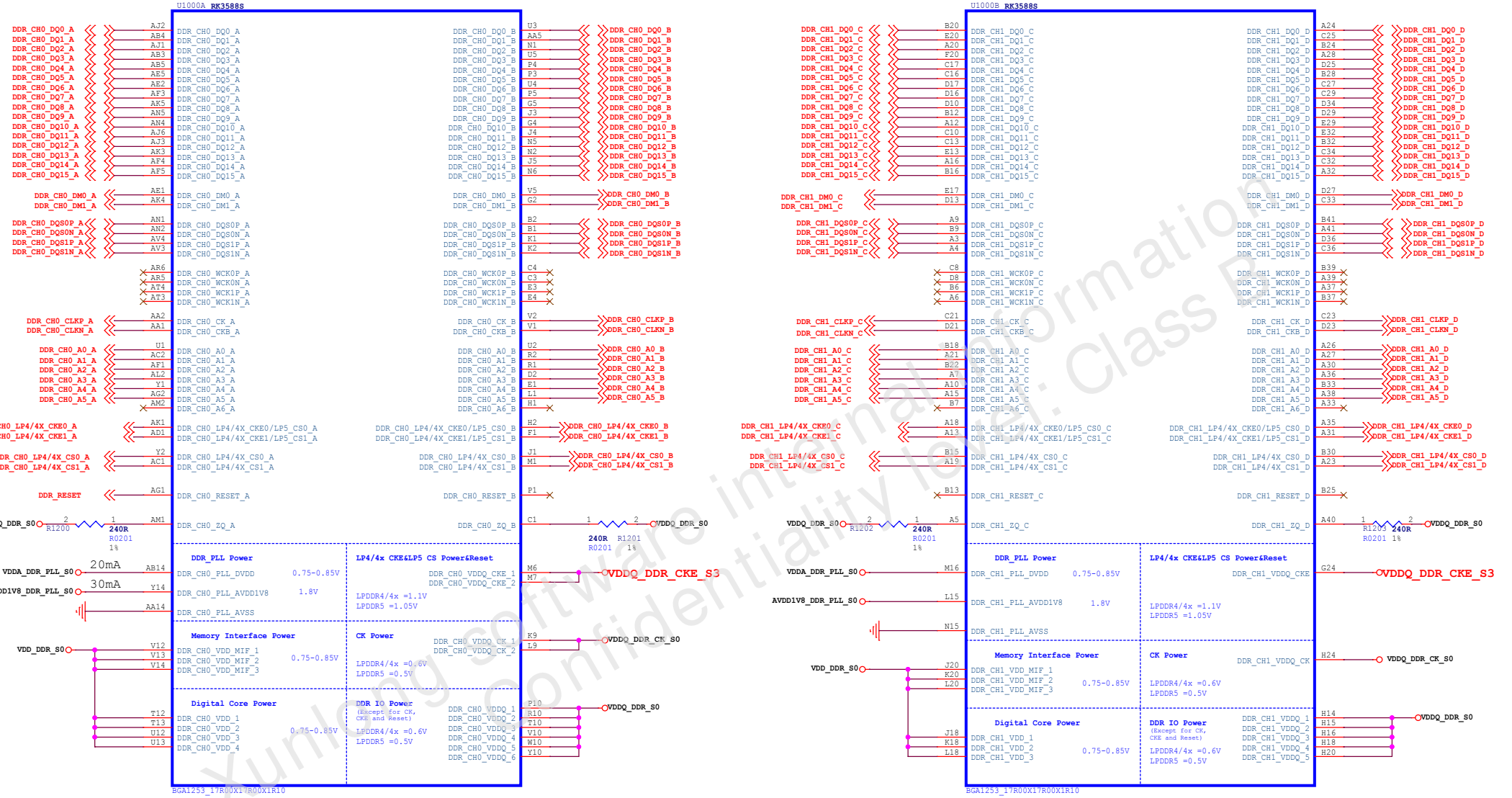
RK3588S(PMUIO2)



Xunlong software internal information
Confidentiality level: class B

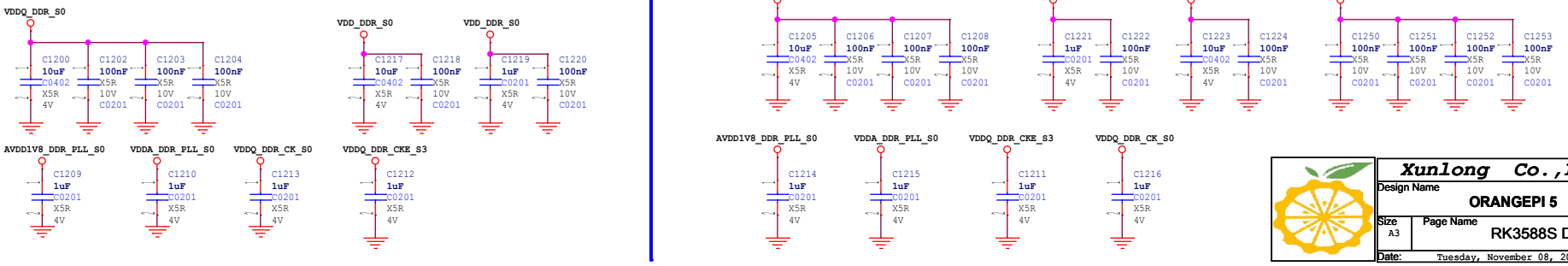
Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: RK3588S_OSC/PLL/PMUIO	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 11 of 36	

RK3588S(DDR PHY)



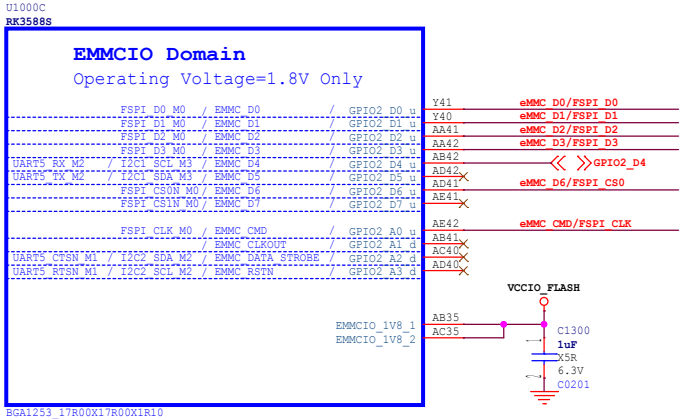
Xunlong software internal information
Confidentiality level: Class B

DDR FILTER



Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: RK3588S DDR Controler	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 12 of 36	

RK3588S(EMMCIO Domain)





RK3588S(VCCIO2 Domain)

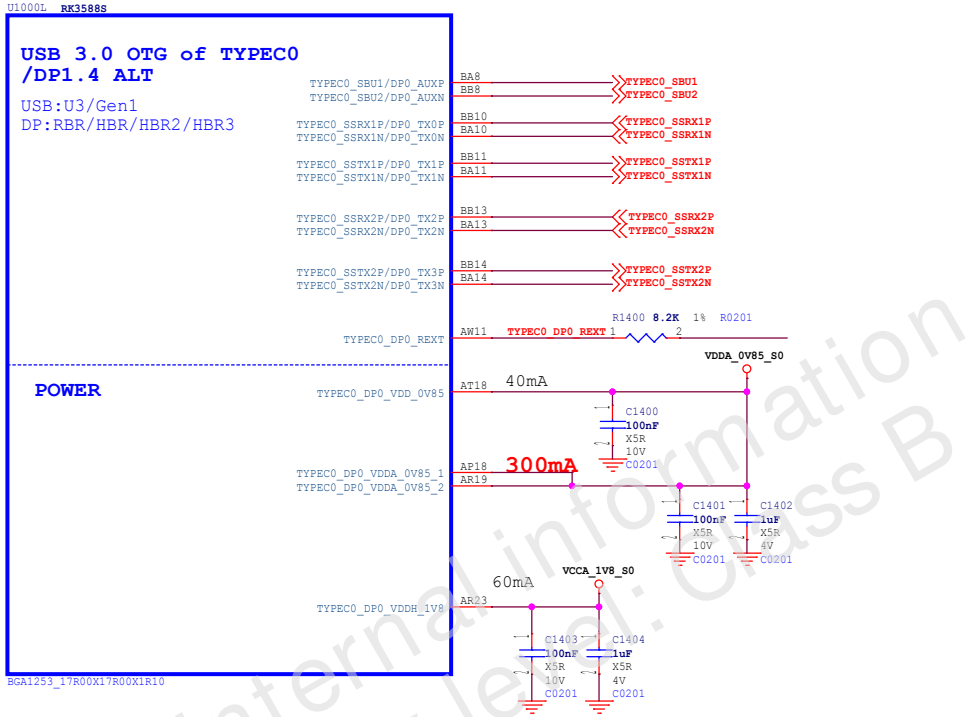




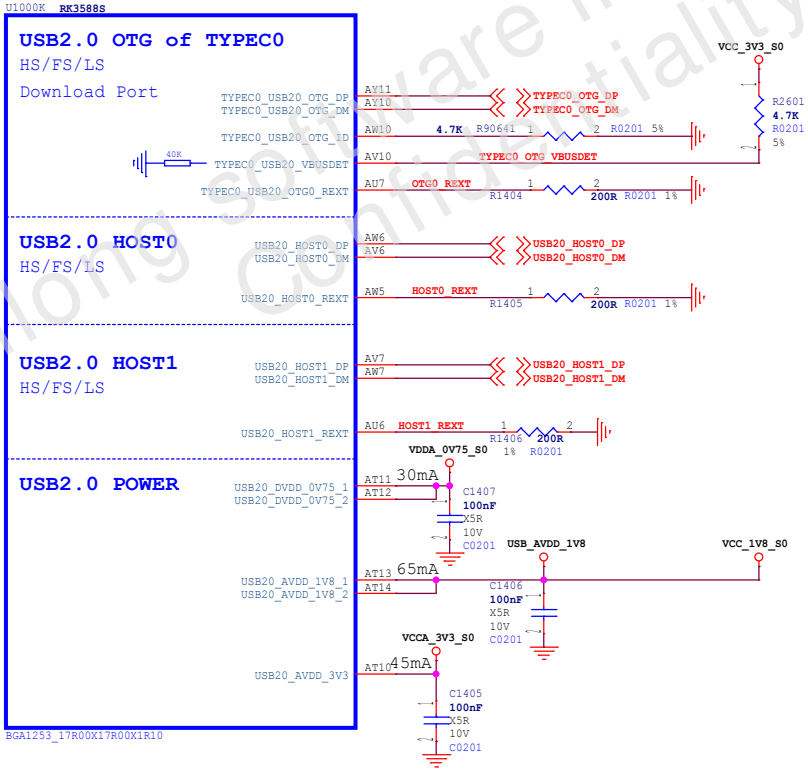
Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: RK3588S Flash/SD Controller	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 13 of 36	

RK3588S(USB3.0/DP1.4)



RK3588S(USB2.0)

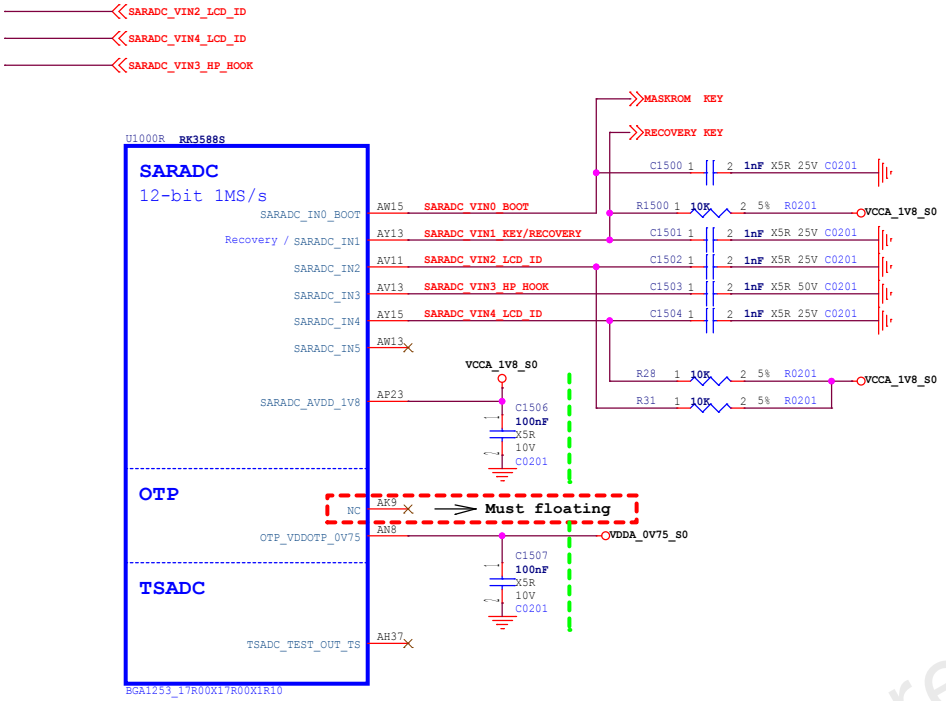


USB Detection

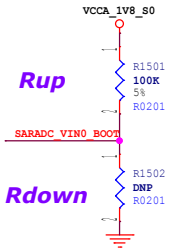
Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: RK3588S_USB20/USB30/DP PHY	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 14 of 36	

RK3588S(SARADC/OTP/TSADC)



BOOT MODE CONFIG

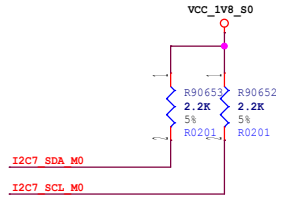


Item	Rup	Rdown	ADC	BOOT MODE(saradc_in5)
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	20K	682	SD Card-USB
LEVEL3	100K	51K	1365	EMMC-USB
LEVEL4	100K	100K	2047	FSPI M0-USB
LEVEL5	100K	200K	2730	FSPI M1-USB
LEVEL6	100K	499K	3412	FSPI M2-USB
LEVEL7	100K	DNP	4095	FSPI_M2-FSPI_M0-EMMC -SD Card-USB

Xunlong software internal information
Confidentiality level: Class B

RK3588S(VCCIO1 Domain)





Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: RK3588S_SARADC/1.8V GPIO	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 15 of 36	

RK3588S(MIPI_CSI)

U1000N

MIPI DPHY CSI_RX Port0

MIPI V1.2
2.5Gbps

MIPI_CSI0_CLK0P / MIPI_CSI0_CLK0N — AN42 / AN41 — MIPI_CSI0_RX_CLK0P / MIPI_CSI0_RX_CLK0N
MIPI_CSI0_D0P / MIPI_CSI0_D0N — AL42 / AL41 — MIPI_CSI0_RX_D0P / MIPI_CSI0_RX_D0N
MIPI_CSI0_D1P / MIPI_CSI0_D1N — AK41 / AK42 — MIPI_CSI0_RX_D1P / MIPI_CSI0_RX_D1N
MIPI_CSI0_CLK1P / MIPI_CSI0_CLK1N — AU41 / AU42
MIPI_CSI0_D2P / MIPI_CSI0_D2N — AT41 / AT42 — MIPI_CSI0_RX_D2P / MIPI_CSI0_RX_D2N
MIPI_CSI0_D3P / MIPI_CSI0_D3N — AP42 / AP41 — MIPI_CSI0_RX_D3P / MIPI_CSI0_RX_D3N

MIPI_CSI0_AVCC0V75 — AM37 — VDDA_0V75_S0
C1602 100nF X5R 10V C0201
C1603 1uF X5R 4V C0201

MIPI_CSI0_AVCC1V8 — AM35 — VCCA_1V8_S0
C1600 1uF X5R 4V C0201
C1601 100nF X5R 10V C0201

RK3588S
BGA1253_17R00X17R00X1R10

MIPI CSI_RX:
100 Ohm ±10%

Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0 MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

RK3588S(MIPI_D/C PHY)

U1000O

MIPI D/C-PHY DSI_TX Port0

D-PHY:V2.0
4.5Gbps/Lane

C-PHY:V1.1
5.7Gbps/Trio

MIPI_DPHY0_TX_CLKP/MIPI_CPHY0_TX_TRIO1_C / MIPI_DPHY0_TX_CLKN/MIPI_CPHY0_TX_TRIO1_B — BB34 / BA34 — MIPI_DPHY0_TX_CLKP / MIPI_DPHY0_TX_CLKN
MIPI_DPHY0_TX_D0P/MIPI_CPHY0_TX_TRIO0_B / MIPI_DPHY0_TX_D0N/MIPI_CPHY0_TX_TRIO0_A — BB31 / BA31 — MIPI_DPHY0_TX_D0P / MIPI_DPHY0_TX_D0N
MIPI_DPHY0_TX_D1P/MIPI_CPHY0_TX_TRIO1_A / MIPI_DPHY0_TX_D1N/MIPI_CPHY0_TX_TRIO0_C — BA32 / BB32 — MIPI_DPHY0_TX_D1P / MIPI_DPHY0_TX_D1N
MIPI_DPHY0_TX_D2P/MIPI_CPHY0_TX_TRIO2_B / MIPI_DPHY0_TX_D2N/MIPI_CPHY0_TX_TRIO2_A — BA35 / BB35 — MIPI_DPHY0_TX_D2P / MIPI_DPHY0_TX_D2N
MIPI_DPHY0_TX_D3P/NO_USE / MIPI_DPHY0_TX_D3N/MIPI_CPHY0_TX_TRIO2_C — BB37 / BA37 — MIPI_DPHY0_TX_D3P / MIPI_DPHY0_TX_D3N

MIPI D/C-PHY CSI_RX Port0

D-PHY:V2.0
4.5Gbps/Lane

C-PHY:V1.1
5.7Gbps/Trio

MIPI_DPHY0_RX_CLKP/MIPI_CPHY0_RX_TRIO1_C / MIPI_DPHY0_RX_CLKN/MIPI_CPHY0_RX_TRIO1_B — BB41 / BA41 — MIPI_DPHY0_RX_CLKP / MIPI_DPHY0_RX_CLKN
MIPI_DPHY0_RX_D0P/MIPI_CPHY0_RX_TRIO0_B / MIPI_DPHY0_RX_D0N/MIPI_CPHY0_RX_TRIO0_A — BA38 / BB38 — MIPI_DPHY0_RX_D0P / MIPI_DPHY0_RX_D0N
MIPI_DPHY0_RX_D1P/MIPI_CPHY0_RX_TRIO1_A / MIPI_DPHY0_RX_D1N/MIPI_CPHY0_RX_TRIO0_C — BA40 / AY40 — MIPI_DPHY0_RX_D1P / MIPI_DPHY0_RX_D1N
MIPI_DPHY0_RX_D2P/MIPI_CPHY0_RX_TRIO2_B / MIPI_DPHY0_RX_D2N/MIPI_CPHY0_RX_TRIO2_A — AY42 / BA42 — MIPI_DPHY0_RX_D2P / MIPI_DPHY0_RX_D2N
MIPI_DPHY0_RX_D3P/NO_USE / MIPI_DPHY0_RX_D3N/MIPI_CPHY0_RX_TRIO2_C — AW41 / AW42 — MIPI_DPHY0_RX_D3P / MIPI_DPHY0_RX_D3N

Power

MIPI_D/C_PHY0_VREG — AT33 MIPI_D/C_PHY0_VREG — C1604 1uF X5R 4V C0201
MIPI_D/C_PHY0_VDD — AR33 — VDDA_0V75_S0
C1605 1uF X5R 4V C0201
C1611 100nF X5R 10V C0201
MIPI_D/C_PHY_VDD_1V2_1 — AR34 — VDDA_1V2_S0
C1607 100nF X5R 10V C0201
MIPI_D/C_PHY_VDD_1V8_1 — AR30 — VCCA_1V8_S0
C1609 100nF X5R 10V C0201

RK3588S
BGA1253_17R00X17R00X1R10

MIPI DPHY:
100 Ohm ±10%
MIPI CPHY:
50 Ohm ±10%

U1000P

MIPI D/C-PHY DSI_TX Port1

D-PHY:V2.0
4.5Gbps/Lane

C-PHY:V1.1
5.7Gbps/Trio

MIPI_DPHY1_TX_CLKP/MIPI_CPHY1_TX_TRIO1_C / MIPI_DPHY1_TX_CLKN/MIPI_CPHY1_TX_TRIO1_B — BB19 / BA19 — MIPI_DPHY1_TX_CLKP / MIPI_DPHY1_TX_CLKN
MIPI_DPHY1_TX_D0P/MIPI_CPHY1_TX_TRIO0_B / MIPI_DPHY1_TX_D0N/MIPI_CPHY1_TX_TRIO0_A — BB16 / BA16 — MIPI_DPHY1_TX_D0P / MIPI_DPHY1_TX_D0N
MIPI_DPHY1_TX_D1P/MIPI_CPHY1_TX_TRIO1_A / MIPI_DPHY1_TX_D1N/MIPI_CPHY1_TX_TRIO0_C — BA17 / BB17 — MIPI_DPHY1_TX_D1P / MIPI_DPHY1_TX_D1N
MIPI_DPHY1_TX_D2P/MIPI_CPHY1_TX_TRIO2_B / MIPI_DPHY1_TX_D2N/MIPI_CPHY1_TX_TRIO2_A — BA20 / BB20 — MIPI_DPHY1_TX_D2P / MIPI_DPHY1_TX_D2N
MIPI_DPHY1_TX_D3P/NO_USE / MIPI_DPHY1_TX_D3N/MIPI_CPHY1_TX_TRIO2_C — BB22 / BA22 — MIPI_DPHY1_TX_D3P / MIPI_DPHY1_TX_D3N

MIPI D/C-PHY CSI_RX Port1

D-PHY:V2.0
4.5Gbps/Lane

C-PHY:V1.1
5.7Gbps/Trio

MIPI_DPHY1_RX_CLKP/MIPI_CPHY1_RX_TRIO1_C / MIPI_DPHY1_RX_CLKN/MIPI_CPHY1_RX_TRIO1_B — BA26 / BB26 — MIPI_DPHY1_RX_CLKP / MIPI_DPHY1_RX_CLKN
MIPI_DPHY1_RX_D0P/MIPI_CPHY1_RX_TRIO0_B / MIPI_DPHY1_RX_D0N/MIPI_CPHY1_RX_TRIO0_A — BA23 / BB23 — MIPI_DPHY1_RX_D0P / MIPI_DPHY1_RX_D0N
MIPI_DPHY1_RX_D1P/MIPI_CPHY1_RX_TRIO1_A / MIPI_DPHY1_RX_D1N/MIPI_CPHY1_RX_TRIO0_C — BB25 / BA25 — MIPI_DPHY1_RX_D1P / MIPI_DPHY1_RX_D1N
MIPI_DPHY1_RX_D2P/MIPI_CPHY1_RX_TRIO2_B / MIPI_DPHY1_RX_D2N/MIPI_CPHY1_RX_TRIO2_A — BB28 / BA28 — MIPI_DPHY1_RX_D2P / MIPI_DPHY1_RX_D2N
MIPI_DPHY1_RX_D3P/NO_USE / MIPI_DPHY1_RX_D3N/MIPI_CPHY1_RX_TRIO2_C — BA29 / BB29 — MIPI_DPHY1_RX_D3P / MIPI_DPHY1_RX_D3N

Power

C1612 1uF X5R 4V C0201
MIPI_D/C_PHY1_VREG — AT27 MIPI_D/C_PHY1_VREG
MIPI_D/C_PHY1_VDD — AR27 — 130mA — VDDA_0V75_S0
C1613 1uF X5R 4V C0201
C1614 100nF X5R 10V C0201
MIPI_D/C_PHY_VDD_1V2_2 — AR35 — 4mA — VDDA_1V2_S0
C1608 100nF X5R 10V C0201
MIPI_D/C_PHY_VDD_1V8_2 — AT30 — 35mA — VCCA_1V8_S0
C1610 100nF X5R 10V C0201

RK3588S
BGA1253_17R00X17R00X1R10

Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited	
Design Name	ORANGEPI 5
Size A3 / Page Name	RK3588S_MIPI Interface / Rev 1.2
Date: Tuesday, November 08, 2022	Sheet 16 of 36

RK3588S(HDMI2.1 TX/eDP1.3 TX)

Note:
The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.

HDMI TX
100 Ohm ±10%

eDP TX
100 Ohm ±10%

HDMI TX
100 Ohm ±10%

U1000Q

HDMI TX/eDP1.3 MUX Port0

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

Option2:HDMI_TX0

HDMI_TX0_D0P/EDP_TX0_D0P	BB4	EDP_TX0_D0P/HDMI0_TX0P	HDMI0_TX0P
HDMI_TX0_D0N/EDP_TX0_D0N	BA4	EDP_TX0_D0N/HDMI0_TX0N	HDMI0_TX0N
HDMI_TX0_D1P/EDP_TX0_D1P	BA5	EDP_TX0_D1P/HDMI0_TX1P	HDMI0_TX1P
HDMI_TX0_D1N/EDP_TX0_D1N	BB5	EDP_TX0_D1N/HDMI0_TX1N	HDMI0_TX1N
HDMI_TX0_D2P/EDP_TX0_D2P	BB7	EDP_TX0_D2P/HDMI0_TX2P	HDMI0_TX2P
HDMI_TX0_D2N/EDP_TX0_D2N	BA7	EDP_TX0_D2N/HDMI0_TX2N	HDMI0_TX2N
HDMI_TX0_D3P/EDP_TX0_D3P	BA2	EDP_TX0_D3P/HDMI0_TX3P	HDMI0_TX3P
HDMI_TX0_D3N/EDP_TX0_D3N	BB2	EDP_TX0_D3N/HDMI0_TX3N	HDMI0_TX3N
HDMI_TX0_SBDP/EDP_TX0_AUXP	BA1	EDP_TX0_AUXP/HDMI0_TX_SBDP	HDMI0_TX_SBDP
HDMI_TX0_SBDN/EDP_TX0_AUXN	AY1	EDP_TX0_AUXN/HDMI0_TX_SBDN	HDMI0_TX_SBDN

HDMI/eDP_TX0_REXT — AY3 HDMI/eDP_TX0_REXT — R1708 1 2 8.2K 1% R0201 — VDDA_0V75_S0

POWER

HDMI/EDP_TX0_VDD_0V75_1 / HDMI/EDP_TX0_VDD_0V75_2 — AM13 / AN12 — 440mA
C1710 100nF X5R 10V C0201
C1711 4.7uF X5R 6.3V C0402

HDMI/EDP_TX0_AVDD_0V75 — AN10 — 1mA
C1712 100nF X5R 10V C0201
C1725 1uF X5R 4V C0201

VCCA_1V8_S0

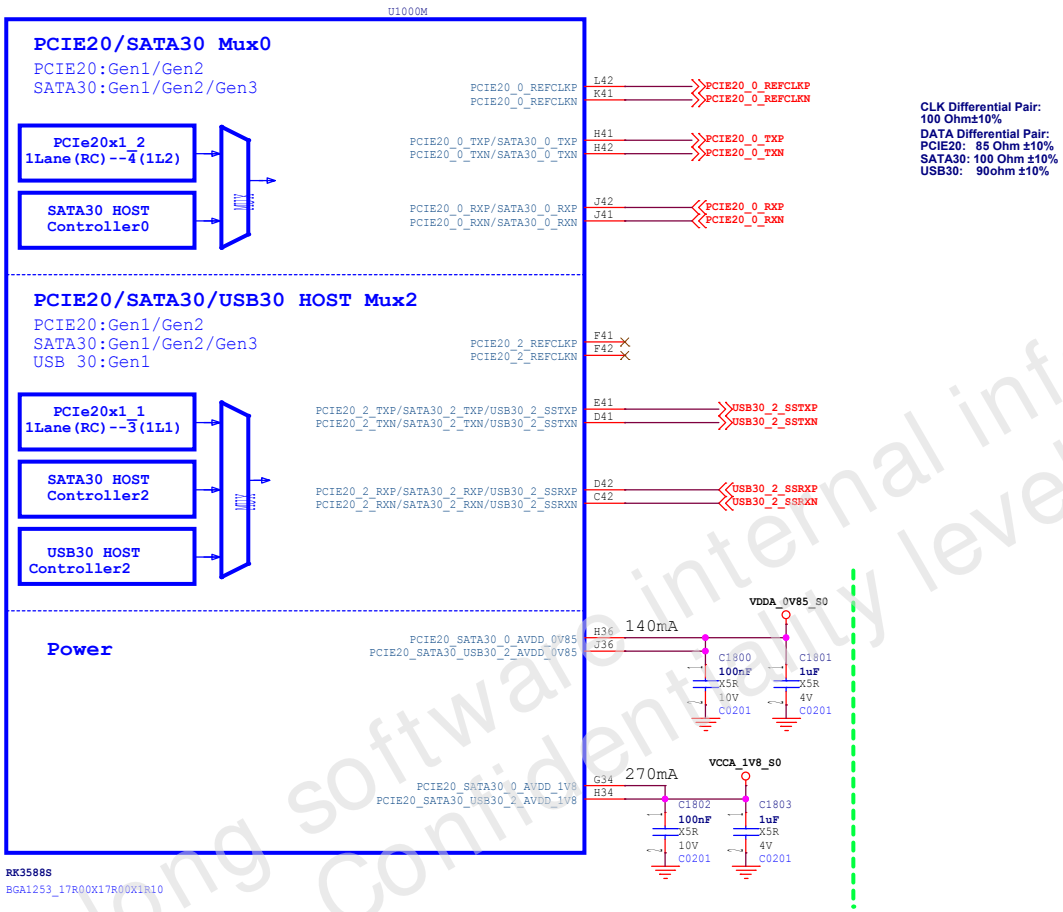
HDMI/EDP_TX0_VDD_IO_1V8 / HDMI/EDP_TX0_VDD_CMN_1V8 — AK12 / AL14 — 200mA
C1713 100nF X5R 10V C0201
C1714 4.7uF X5R 6.3V C0402

RK3588S
BGA1253_17R00X17R00X1R10

Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A4	Page Name: RK3588S_HDMI/eDP Interface	Rev: 1.2
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RK3588S(PCIE20/SATA30/USB30)



PCIe2.0 PHY

Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ_M* PCIE20X1_1_WAKEN_M* PCIE20X1_1_PERSTN_M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ_M* PCIE20X1_2_WAKEN_M* PCIE20X1_2_PERSTN_M* PCIE20X1_2_BUTTON_RSTN

Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited

Design Name: ORANGEPI 5

Size: A3 | Page Name: RK3588S PCIE2/SATA3/USB3 PHY | Rev: 1.2

Date: Tuesday, November 08, 2022 | Sheet 18 of 36

RK3588S(VCCIO4 Domain)

U1000H

VCCIO4 Domain
Operating Voltage=1.8V/3.3V

PCIE20X1_1_CLKREQN_M2 / SPI0_HPDIN_M2					
PCIE20X1_1_WAKEN_M2	UART6_RX_M1	SPI4_MISO_M2	I2C2_SDA_M4	GPIO1_A0_d	
	UART6_TX_M1	SPI4_MOSI_M2	I2C2_SCL_M4	GPIO1_A1_d	
PWM0_M2	VOP_POST_EMPTY	UART6_RTSN_M1	SPI4_CLK_M2	I2C4_SDA_M3	GPIO1_A2_d
PWM1_M2		UART6_CTSN_M1	SPI4_CS0_M2	I2C4_SCL_M3	GPIO1_A3_d
			SPI2_MISO_M0		GPIO1_A4_d
	HDMI_TX0_HPD_M0		SPI2_MOSI_M0		GPIO1_A5_d
			SPI2_CLK_M0		GPIO1_A6_d
PWM3_IR_M3 / PCIE20X1_1_PERSTN_M2			SPI2_CS0_M0	PDM1_SDI0_M1	GPIO1_A7_u
			SPI2_CS1_M0	PDM1_SDI1_M1	GPIO1_B0_u
			SPI0_MISO_M2	PDM1_SDI2_M1	GPIO1_B1_d
		UART4_RX_M2	SPI0_MOSI_M2	PDM1_SDI3_M1	GPIO1_B2_d
	SATA0_ACT_LED_M1	UART4_TX_M2	SPI0_CLK_M2	PDM1_CLK1_M1	GPIO1_B3_d
		UART7_RX_M2	SPI0_CS0_M2	PDM1_CLK0_M1	GPIO1_B4_u
		UART7_TX_M2	SPI0_CS1_M2		GPIO1_B5_u
PWM13_M2 / SATA2_ACT_LED_M1	MIPI_CAMERA1_CLK_M0	UART1_TX_M1	SPDIF0_TX_M0	I2C5_SCL_M3	GPIO1_B6_u
	MIPI_CAMERA2_CLK_M0	UART1_RX_M1	SPDIF1_TX_M0	I2C5_SDA_M3	GPIO1_B7_u
PWM14_M2	MIPI_CAMERA3_CLK_M0 /	UART1_RTSN_M1		I2C8_SCL_M2	GPIO1_D6_u
PWM15_IR_M3	MIPI_CAMERA4_CLK_M0 /	UART1_CTSN_M1		I2C8_SDA_M2	GPIO1_D7_u

G40	LCD_MIPI_PWREN_H	
L40	LCD_MIPI_PWREN_H_1	
D38	PWM0_LED	I2C4_SDA_M3_TP
L39	PWM1/GPIO1_A3	I2C4_SCL_M3_TP
G37	TP1_RST_L	
M40	HDMI_TX0_HPD_M0	
D39	MIPI_CAM2_RST_L	
H38	TP1_INT_L	
H39	MIPI_TE1	
G39	LCD_MIPI_RESET_0	
M38	MIPI_CAM1_RST_L	
M37	LCD_MIPI_RESET_1	
M39	TP_RST_L	
D40	TP_INT_L	
L38	I2C5_SCL/UART1_TX/GPIO1_B6	
E37	I2C5_SDA/UART1_RX/GPIO1_B7	
L37	MIPI_CAM3_CLKOUT	
G38	MIPI_CAM4_CLKOUT	

VCC_1V8_S0
VCCIO4_1V8_1 G27
VCCIO4_1V8_2 G28
C1902 100nF X5R 10V C0201
VCC_1V8_S0
VCCIO4 G31
C1903 100nF X5R 10V C0201

RK3588S
BGA1253_17R00X17R00X1R10

RK3588S(VCCIO6 Domain)

U1000J

VCCIO6 Domain
Operating Voltage=1.8V/3.3V

BT1120_D0 / CIF_D0		I2S1_MCLK_M0	UART9_RTSN_M1	SPI0_MISO_M1	PCIE20X1_1_CLKREQN_M1 / GPIO4_A0_d
BT1120_D1 / CIF_D1		I2S1_SCLK_M0	UART9_CTSN_M1	SPI0_MOSI_M1	PCIE20X1_1_WAKEN_M1 / GPIO4_A1_d
BT1120_D2 / CIF_D2		I2S1_LRCK_M0		SPI0_CLK_M1	PCIE20X1_1_PERSTN_M1 / GPIO4_A2_d
BT1120_D3 / CIF_D3			UART0_TX_M2		GPIO4_A3_d
BT1120_D4 / CIF_D4			UART0_RX_M2	SPI2_MISO_M1 / I2C3_SCL_M2	GPIO4_A4_d
BT1120_D5 / CIF_D5		I2S1_SDI0_M0	UART3_TX_M2	SPI2_MOSI_M1 / I2C3_SDA_M2	GPIO4_A5_d
BT1120_D6 / CIF_D6		I2S1_SDI1_M0	UART3_RX_M2	SPI2_CLK_M1 / I2C5_SCL_M2	GPIO4_A6_d
BT1120_D7 / CIF_D7		I2S1_SDI2_M0		SPI2_CS0_M1 / I2C5_SDA_M2	GPIO4_A7_d
BT1120_CLKOUT / CIF_CLKIN		I2S1_SDI3_M0	UART8_TX_M0	SPI2_CS1_M1 / I2C6_SDA_M3	GPIO4_B0_d
SPDIF1_TX_M1	MIPI_CAMERA0_CLK_M0 / SATA2_ACT_LED_M0 /	I2S1_SDO0_M0	UART8_RX_M0	SPI0_CS1_M1 / I2C6_SCL_M3	GPIO4_B1_u
CAN1_RX_M1 / BT1120_D8 / CIF_HREF		PWM14_M1	I2S1_SDO1_M0	UART8_RTSN_M0	SPI0_CS0_M1 / I2C7_SCL_M3 / PCIE20X1_1_BUTTON_RSTN / GPIO4_B2_u
CAN1_TX_M1 / BT1120_D9 / CIF_VSYNC		PWM15_IR_M1	I2S1_SDO2_M0	UART8_CTSN_M0	I2C7_SDA_M3 / PCIE20X1_2_BUTTON_RSTN / GPIO4_B3_u
BT1120_D10 / CIF_CLKOUT		PWM11_IR_M1	I2S1_SDO3_M0	UART9_TX_M1	SPI0_HPDIN_M0 / SPDIF0_TX_M1 / GPIO4_B4_u
BT1120_D11 /		PWM12_M1		UART9_RX_M1	SPI3_MISO_M1 / GPIO4_B5_d
BT1120_D12 /		PWM13_M1			SPI3_MOSI_M1 / I2C5_SCL_M1 / SATA0_ACT_LED_M0 / GPIO4_B6_d
BT1120_D13 /		HDMI_TX0_SCL_M0			SPI3_CLK_M1 / I2C5_SDA_M1 / PCIE20X1_2_CLKREQN_M1 / GPIO4_B7_u
BT1120_D14 /		HDMI_TX0_SDA_M0			SPI3_CS0_M1 / I2C8_SCL_M3 / PCIE20X1_2_WAKEN_M1 / GPIO4_C0_u
BT1120_D15 / SPDIF1_TX_M2		PWM6_M1	HDMI_TX0_CEC_M0		SPI3_CS1_M1 / I2C8_SDA_M3 / PCIE20X1_2_PERSTN_M1 / GPIO4_C1_d

AV19	I2S1_MCLK
AW18	I2S1_SCLK_TX
AV26	I2S1_LRCK_TX
AV19	UART0_TX_M2/GPIO4_A3
AW19	UART0_RX_M2/GPIO4_A4
AU15	TYPEC0_SBU1_DC
AV18	I2S1_SDI1
AW26	TYPEC0_SBU2_DC
AW27	I2C6_SDA_M3
AU22	I2C6_SCL_M3
AT15	CAN1_RX_M1/GPIO4_B2
AV23	CAN1_TX_M1/GPIO4_B3
AV27	I2S1_SDO3
AU23	SDMMC_PWREN
AW22	HDMI0_TX_ON_H
AV22	HDMI_TX0_SCL_M0
AW23	HDMI_TX0_SDA_M0
AV26	HDMI_TX0_CEC_M0

Note:
BT1120 Only Support Output

VCC_1V8_S0
VCCIO6_1V8 AJ34
C1906 100nF X5R 10V C0201
VCC_3V3_S0
VCCIO6_1 AL33
VCCIO6_2 AM33
C1907 100nF X5R 10V C0201

RK3588S
BGA1253_17R00X17R00X1R10

RK3588S(VCCIO5 Domain)

U1000I

VCCIO5 Domain
Operating Voltage=1.8V/3.3V

PWM10_M0	SPI4_MISO_M1 / I2C6_SDA_M4 / GMAC1_TXD2	I2S3_MCLK	FSPI_D0_M2	SDIO_D0_M1	GPIO3_A0_u
AUDDSM_LN / PWM11_IR_M0	SPI4_MOSI_M1 / I2C6_SCL_M4 / GMAC1_TXD3	I2S3_SCLK	FSPI_D1_M2	SDIO_D1_M1	GPIO3_A1_u
AUDDSM_LP / UART8_TX_M1	SPI4_CLK_M1 /	GMAC1_RXD2	I2S3_LRCK	FSPI_D2_M2	SDIO_D2_M1 / GPIO3_A2_u
AUDDSM_RN / UART8_RX_M1	SPI4_CS0_M1 /	GMAC1_RXD3	I2S3_SDO	FSPI_D3_M2	SDIO_D3_M1 / GPIO3_A3_u
AUDDSM_RP / UART8_RTSN_M1	SPI4_CS1_M1 /	GMAC1_TXCLK	I2S3_SDI		SDIO_CMD_M1 / GPIO3_A4_d
UART8_CTSN_M1 /	I2C4_SDA_M0 /	GMAC1_RXCLK	FSPI_CLK_M2	MIPI_CAMERA0_CLK_M1	SDIO_CLK_M1 / GPIO3_A5_d
	I2C4_SCL_M0 /	RTH1_REFCLKO_25M		MIPI_CAMERA1_CLK_M1 /	GPIO3_A6_d
		GMAC1_RXD0		MIPI_CAMERA2_CLK_M1 / PWM8_M0	GPIO3_A7_u
		GMAC1_RXD1		MIPI_CAMERA3_CLK_M1 / PWM9_M0	GPIO3_B0_u
UART2_TX_M2		GMAC1_RXDV_CRS		MIPI_CAMERA4_CLK_M1 / PWM10_M1	GPIO3_B1_d
UART2_RX_M2		GMAC1_TXER	I2S2_SDI_M1		PWM3_IR_M1 / GPIO3_B2_d
UART2_RTSN		GMAC1_TXD0	I2S2_SDO_M1		GPIO3_B3_u
UART2_CTSN		GMAC1_TXD1	I2S2_MCLK_M1		GPIO3_B4_u
UART3_TX_M1	CAN1_RX_M0 /	GMAC1_TXEN	I2S2_SCLK_M1		PWM12_M0 / GPIO3_B5_u
UART3_RX_M1	CAN1_TX_M0 /	GMAC1_MCLKINOUT	I2S2_LRCK_M1		PWM13_M0 / GPIO3_B6_d
	SPI1_MOSI_M1 / I2C3_SCL_M1 /	GMAC1_PTP_REF_CLK /			GPIO3_B7_d
UART7_TX_M1	SPI1_MISO_M1 / I2C3_SDA_M1 /	GMAC1_PPSTRIG			GPIO3_C0_d
UART7_RX_M1	SPI1_CLK_M1 /	GMAC1_PPSOUT			GPIO3_C1_d
UART7_RTSN_M1	SPI1_CS0_M1 / I2C8_SCL_M4 /	GMAC1_MDC		MIPI_TE0	PWM14_M0 / GPIO3_C2_d
UART7_CTSN_M1	SPI1_CS1_M1 / I2C8_SDA_M4 /	GMAC1_MDIO		MIPI_TE1	PWM15_IR_M0 / GPIO3_C3_d
UART5_TX_M1	SPI3_CS0_M3 / CAN2_RX_M0 /	CIF_D8		FSPI_CS0N_M2	GPIO3_C4_u
UART5_RX_M1	SPI3_CS1_M3 / CAN2_TX_M0 /	CIF_D9		FSPI_CS1N_M2	GPIO3_C5_u
	SPI3_MISO_M3 /	CIF_D10			GPIO3_C6_u
	SPI3_MOSI_M3 / I2C5_SCL_M0 /	CIF_D11	PCIE20X1_2_CLKREQN_M0 /	HDMI_TX0_SCL_M2	GPIO3_C7_u
UART4_RX_M1	SPI3_CLK_M3 / I2C5_SDA_M0 /	CIF_D12	PCIE20X1_2_WAKEN_M0 /	HDMI_TX0_SDA_M2	PWM8_M2 / GPIO3_D0_u
UART4_TX_M1	SPI0_MISO_M3 /	CIF_D13	PCIE20X1_2_PERSTN_M0 /		PWM9_M2 / GPIO3_D1_d
UART9_RTSN_M2 / SPI0_MOSI_M3 /	I2C7_SCL_M2 /	CIF_D14			GPIO3_D2_d
UART9_CTSN_M2 / SPI0_CLK_M3 /	I2C7_SDA_M2 /	CIF_D15			PWM10_M2 / GPIO3_D3_d
UART9_RX_M2 / SPI0_CS0_M3 /		MCU_JTAG_TCK_M1		HDMI_TX0_HPD_M1	GPIO3_D4_d
UART9_TX_M2 / SPI0_CS1_M3 /		MCU_JTAG_TMS_M1			PWM11_IR_M3 / GPIO3_D5_d

AR38	GMAC0_TXD2	
AR37	GMAC0_TXD3	
AT38	GMAC0_RXD2	
AT40	GMAC0_RXD3	
AT39	GMAC0_TXCLK	
AV38	GMAC0_RXCLK	
AV37	BT_REG_ON_H	GPIO3_A6
AT37	GMAC0_RXD0	
AT39	GMAC0_RXD1	
AV39	GMAC0_RXDV_CRS	
AW34	GMAC0_RSTn_L	
AW35	GMAC0_TXD0	
AV35	GMAC0_TXD1	
AY35	GMAC0_TXEN	
AW37	GMAC0_MCLKINOUT	
AY34	MIPI_TE0	
AR36	TYPEC_EN	MAC_INTB
AW38	MIPI_CAM1_PDN_L	
AV40	GMAC0_MDC	
AW39	GMAC0_MDIO	
AU34	MIPI_CAM3_RST_L	
AV34	MIPI_CAM2_PDN_L	
AV30	MIPI_CAM3_PDN_L	
AU30	PCIE20x1_2_CLKREQn_M0	
AW31	PCIE20x1_2_WAKEn_M0	
AY27	PCIE20x1_2_PERSTn_M0	
AY30	UART9_RTSN_M2_BT	
AY31	UART9_CTSN_M2_BT	
AV31	UART9_RX_M2_BT	
AW30	UART9_TX_M2_BT	

RESERVED AF35
VCCIO5_1V8 AF36
VCC_1V8_S0
C1904 100nF X5R 10V C0201
VCCIO5_3 AC33
VCCIO5_2 AC34
VCC_1V8_S0
C1905 100nF X5R 10V C0201

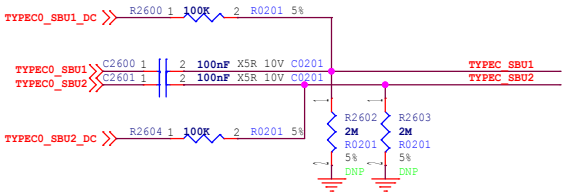
RK3588S
BGA1253_17R00X17R00X1R10

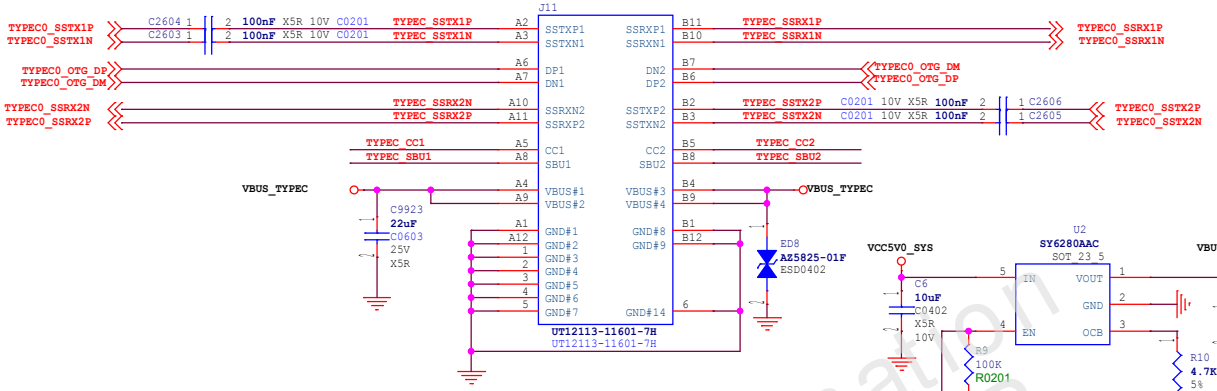
VCC_3V3_S0
R1914 2.2K 5% R0201
R1915 2.2K 5% R0201
I2C6_SCL_M3
I2C6_SDA_M3

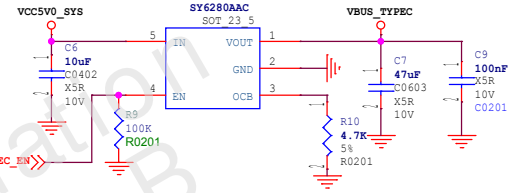
Xunlong internal information
Confidentiality level: Class

Xunlong Co.,Limited			
Design Name: ORANGEPI 5			
Size: A2	Page Name: RK3588S GPIO		Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 19 of 36		

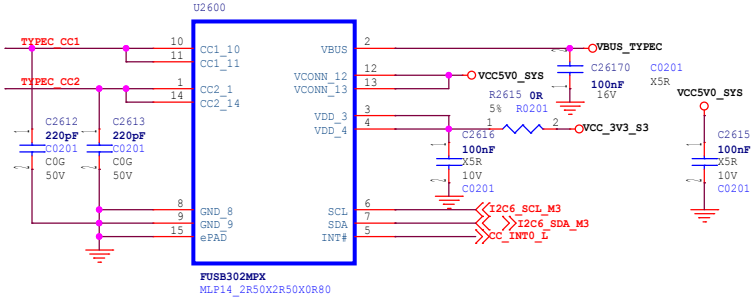
USB Type-C Port

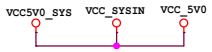






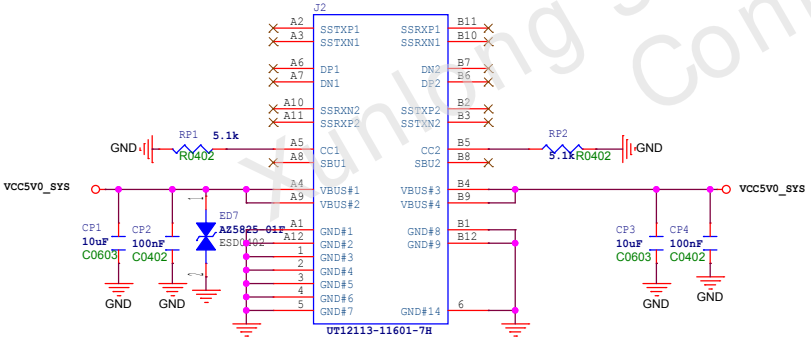
USB Type-C CC CTRL



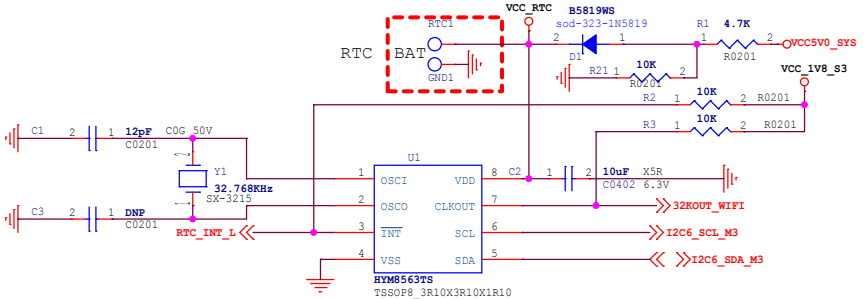


DCIN-5V/4A

TPYE-C POWER IN



RTC IC



Address:Read A3H,Write A2H

Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: Power_Type-C	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 20 of 36	

PMIC1 RK806-1

SLEEP1_RESET1_DEVOFF1_VSEL1
VSEL4
VSEL5
PMIC_INT_L
PMIC_RESET_L
PMIC_SPI_CLK
PMIC_SPI_CS
PMIC_SPI_MOSI

VCC_SYSIN
RD2200 AZ5825-01F RSD0402-6

IF TVS UNMOUNTED, ESD OR SURGE SHOULD BE DAMAGE THE PMIC!!!

This device must be mounted.Replacing TVS mode is not recommended, if must, please choose the same specifications
Operating Supply Vltage : 5.5V(5.25-6V)
PeakPulse Current: >10A (tp=8/20uS)
Surge Clamping Voltage: <6.5V

DO NOT DELETE IT!

VSEL5 — R2242 0R — R0201 5% — PMIC_PWR_CTRL2
R2244 DNP 120K R0201

VSEL4 — R2290 0R — R0201 5% — PMIC_PWR_CTRL3
R2291 DNP 120K R0201

PMIC RK806-1 BUCK

U2200A

BUCK5 2.5A
峰值电流 2.5A
VDD_DDR_S0 0.80V
Default:0.85V
Low frequency: 0.85V-->0.75V
0.675 0.85 0.935
C2200 22uF X5R C0603
C2202 47uF X5R 10V C0603
C2203 22uF X5R 10V C0603
R2200 100R 5% R0201
C9922 1uF X5R 6.3V C0201
L2200 0.47uH IND_201610
PMIC_SW5
Feedback from RK3588S

BUCK1 6.5A
峰值电流 6A
VCC_SYSIN
C2201 22uF X5R C0603
T85
VDD_GPU_S0 0.85V
0.55 0.75 0.95
L2201 0.22uH IND_404020
C2209 1uF X5R 6.3V C0201
R2208 100R 5% R0201
C2210 22uF X5R 10V C0603
C2205 47uF X5R 10V C0603
C2280 47uF X5R 10V C0603
PMIC_SW1
PMIC_VOUT1
VDD_GPU_S0
Feedback from RK3588S

BUCK6 2.5A
峰值电流 1.0A
VCC_1V1_NLDO_S3 600mA
VDD2_DDR_S3 1.10V 658mA
1.045 1.1 1.155
C2214 47uF X5R 10V C0603
C2215 22uF X5R 10V C0603
C2216 100pF C0G 25V C0201
L2206 0.47uH IND_201610
R2218 120K 1% R0201
R2227 100K 1% R0201
PMIC_SW6
PMIC_VOUT6
PMIC_FB6

LPDDR4/4x=1.1V	120K
LPDDR5=1.05V	110K

BUCK2 5A
峰值电流 3A
VCC_SYSIN
C2213 22uF X5R C0603
T95
VDD_CPU_LIT_S0 0.95V
0.55 0.75 0.95
L2205 0.24uH IND_252012
C2221 1uF X5R 6.3V C0201
R2219 100R 5% R0201
C2217 22uF X5R 10V C0603
C2218 22uF X5R 10V C0603
C2281 22uF X5R 10V C0603
PMIC_SW2
PMIC_VOUT2
VDD_CPU_LIT_S0
Feedback from RK3588S

BUCK9 2.5A
峰值电流 2.5A
VDDQ_DDR_S3 232.60mA 0.60V
0.57 0.6 0.63
0.475 0.5 0.525
C2225 22uF X5R C0603
T60
C2228 47uF X5R 10V C0603
C2227 22uF X5R 10V C0603
C2229 100pF C0G 25V C0201
L2204 0.47uH IND_201610
R2236 20K 1% R0201
R2241 100K 1% R0201
PMIC_SW9
PMIC_VOUT9
PMIC_FB9

LPDDR4/4x=0.6V	20K	100K
LPDDR5=0.5V	0R	DNP

BUCK3 5A
峰值电流 2A
VCC_SYSIN
C2226 22uF X5R C0603
T75
VDD_LOGIC_S0 0.75V
0.675 0.75 0.825
L2205 0.24uH IND_252012
C2230 1uF X5R 6.3V C0201
R2238 100R 5% R0201
C2231 22uF X5R 10V C0603
C2232 47uF X5R 10V C0603
C2282 47uF X5R 10V C0603
PMIC_SW3
VDD_LOGIC_S0
Feedback from RK3588S

BUCK8 2.5A
峰值电流 2.5A
VCC_3V3_S3
T13
C2238 47uF X5R 10V C0603
C2239 22uF X5R 10V C0603
L2206 0.47uH IND_201610
PMIC_SW8
PMIC_VOUT8

BUCK4 5A
峰值电流 2A
VCC_SYSIN
C2236 22uF X5R C0603
T77
VDD_VDENC_S0 0.775V
0.675 0.75 0.825
L2207 0.24uH IND_252012
C2291 1uF X5R 6.3V C0201
R2209 100R 5% R0201
C2240 22uF X5R 10V C0603
C2237 22uF X5R 10V C0603
C2283 47uF X5R 10V C0603
PMIC_SW4
VDD_VDENC_S0
Feedback from RK3588

BUCK10 2.5A
峰值电流 1A
VCC1V8_PMU_DDR_S3
Default:1.8V
VCC_1V8_S3
T18
C2241 22uF X5R C0603
C2242 47uF X5R 10V C0603
C2243 22uF X5R 10V C0603
L2208 0.47uH IND_201610
PMIC_SW10
PMIC_VOUT10

BUCK7 2.5A
峰值电流 0.5A
VCC_SYSIN
T20
VCC_2V0_PLDO_S3 0.35A 2.0V
C2244 22uF X5R 10V C0603
C2245 47uF X5R 10V C0603
L2209 0.47uH IND_201610
PMIC_SW7
PMIC_VOUT7

RK806-1 QFN68_7R00X7R00X0R80_T

Xunlong software internal information
Confidentiality level: Class B

PMIC RK806-1 Managerment

VCC_1V8_S3_PLDO6
U2200C
C2248 1uF 6.3V X5R C0201
VCCIO
(IIC/SPI)CS — PMIC_SPI_CS
(SDA)MOSI — PMIC_SPI_MOSI
(SCL)CLK — PMIC_SPI_CLK
(PWRCTRL3)SO — PMIC_PWR_CTRL3
PWRCTRL2 — PMIC_PWR_CTRL2
PWRCTRL1 — SLEEP1_RESET1_DEVOFF1_VSEL1
R2258 100K 5% R02Q1
VCC_1V8_S3
R2261 10K 5% R0201
R2263 10K 5% R0201
PMIC_INT_L
PMIC_RESET_L
VCCA
VCC_SYSIN
C2254 10uF X5R C0402
SYNC_CLK
SYNC
VDC
PWRON
(M/S)EXT_EN
INT
RESETB
ePAD
RK806-1 QFN68_7R00X7R00X0R80_T
RESET
T1
C2262 100nF X5R 10V C0201
C2261 100nF X5R 10V C0201
20220929
VCC_SYSIN
C4 10uF X5R C0402
R2264 510K R0201
PMIC_PDC
D6 B5819WS sod-323-1N5819
R2265 510K R0201
C2260 1nF X5R 25V C0201
POWER KEY

RK3588S nPOR
TSADC_SHUT
RK806-1

Note:
Reset Key Control Path
TSADC_SHUT Control Path

PMIC RK806-1 LDO

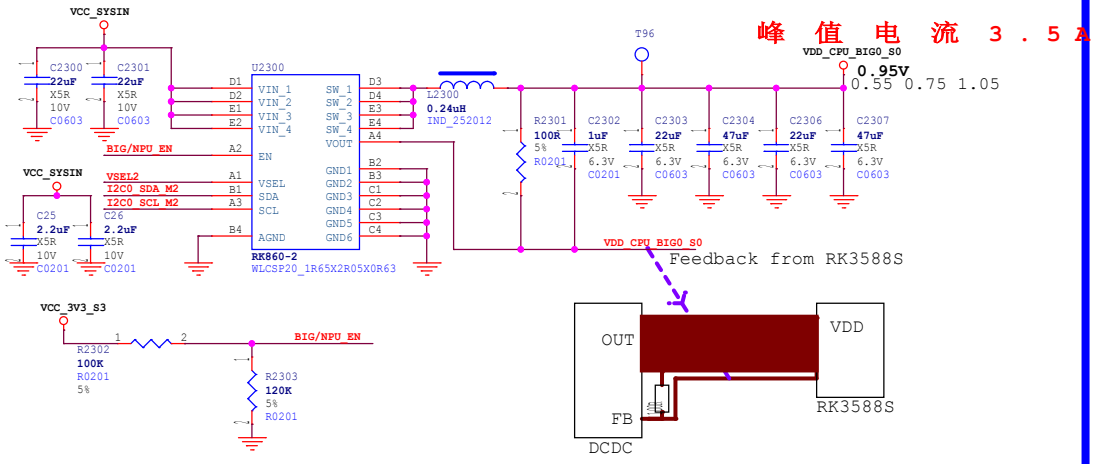
U2200B
VCC_2V0_PLDO_S3
C2247 1uF X5R 6.3V C0201
VCC_SYSIN
C2252 10uF X5R 10V C0402
VCC_1V1_NLDO_S3
C2257 1uF X5R 6.3V C0201
VCC_1V1_NLDO_S3
C2264 1uF X5R 6.3V C0201
VCC11
VCC12
VCC13
VCC14
PLDO
0.5A
0.3A
PLDO1
PLDO2
PLDO3
PLDO4
PLDO5
NLDO1
NLDO2
NLDO3
NLDO4
NLDO5

VCC_1V8_S0 200mA — C2246 1uF — Default:1.8V
VCCA_1V8_S0 150mA — C2249 1uF — Default:1.8V
VDDA_1V2_S0 2mA — C2250 1uF — Default:1.2V
VCCA_3V3_S0 200mA — C2251 1uF — Default:3.3V
VCCIO_SD_S0 1mA — C2253 1uF — Default:3.3V
VDD_0V75_S3 10mA — C2255 2.2uF — Default:0.75V
VDDA_DDR_PLL_S0 126mA — C2258 2.2uF — Default:0.85V; Low frequency:0.85V-->0.75V
VDDA_0V75_S0 500mA — C2259 2.2uF — Default:0.75V
VDDA_0V85_S0 500mA — C2263 2.2uF — Default:0.85V

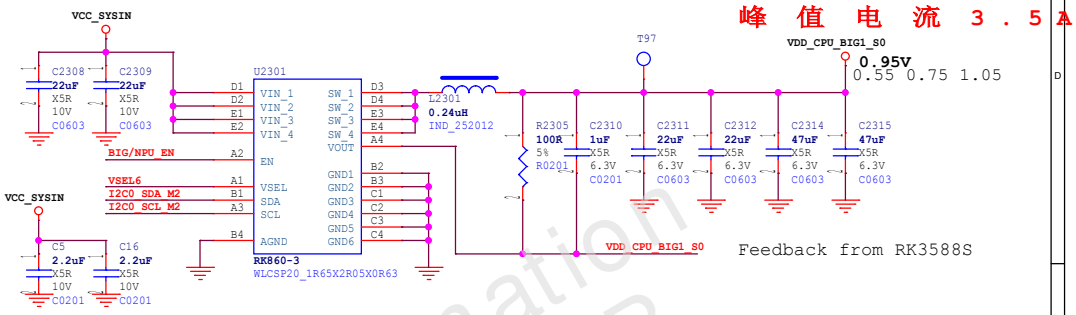
2.7 3.3 3.6
1.65 1.8 1.95
VCCA_1V8_S0
VCC_1V8_S0
AVDD1V8_DDR_PLL_S0
C2290 1uF X5R 6.3V C0201

Xunlong Co.,Limited						
Design Name	ORANGEPI5					
Size	Number	Revision	Date	File	Sheet of	
A3		V1.0	2022-09-29			

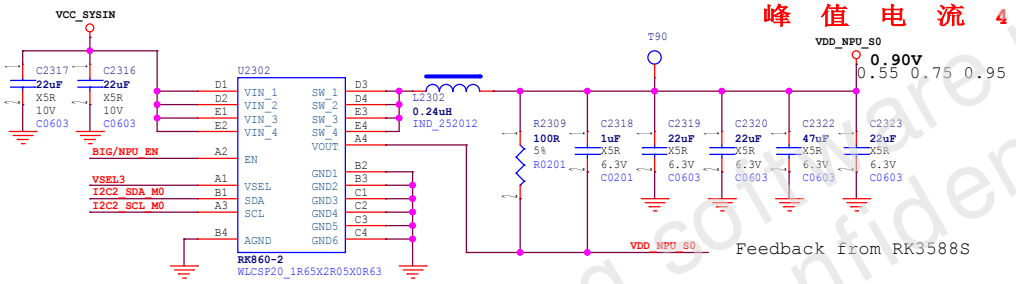
VDD_CPU_BIG0



VDD_CPU_BIG1



VDD_NPU

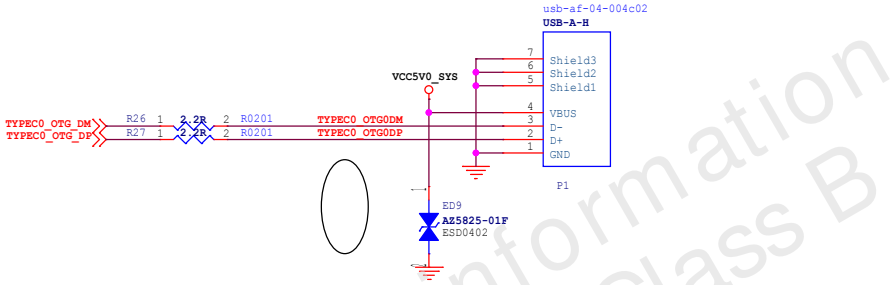


I2C0_SCL_M2
I2C0_SDA_M2
I2C2_SCL_M0
I2C2_SDA_M0
VSEL2
VSEL3
VSEL6

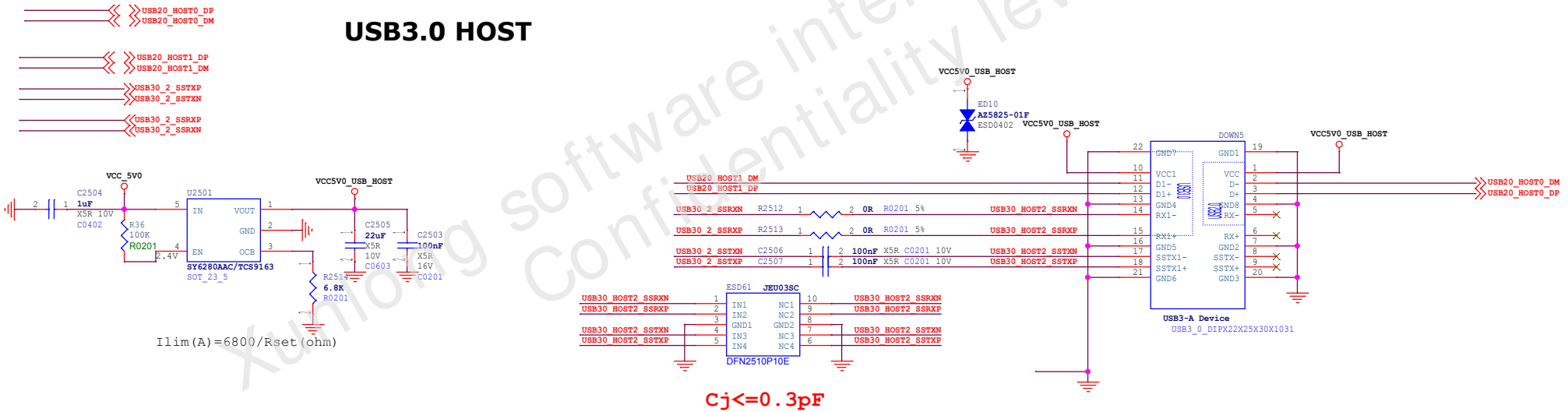
Xunlong software Internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: Power_Ext Discrete	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 22 of 36	

USB2.0 Type-A Port

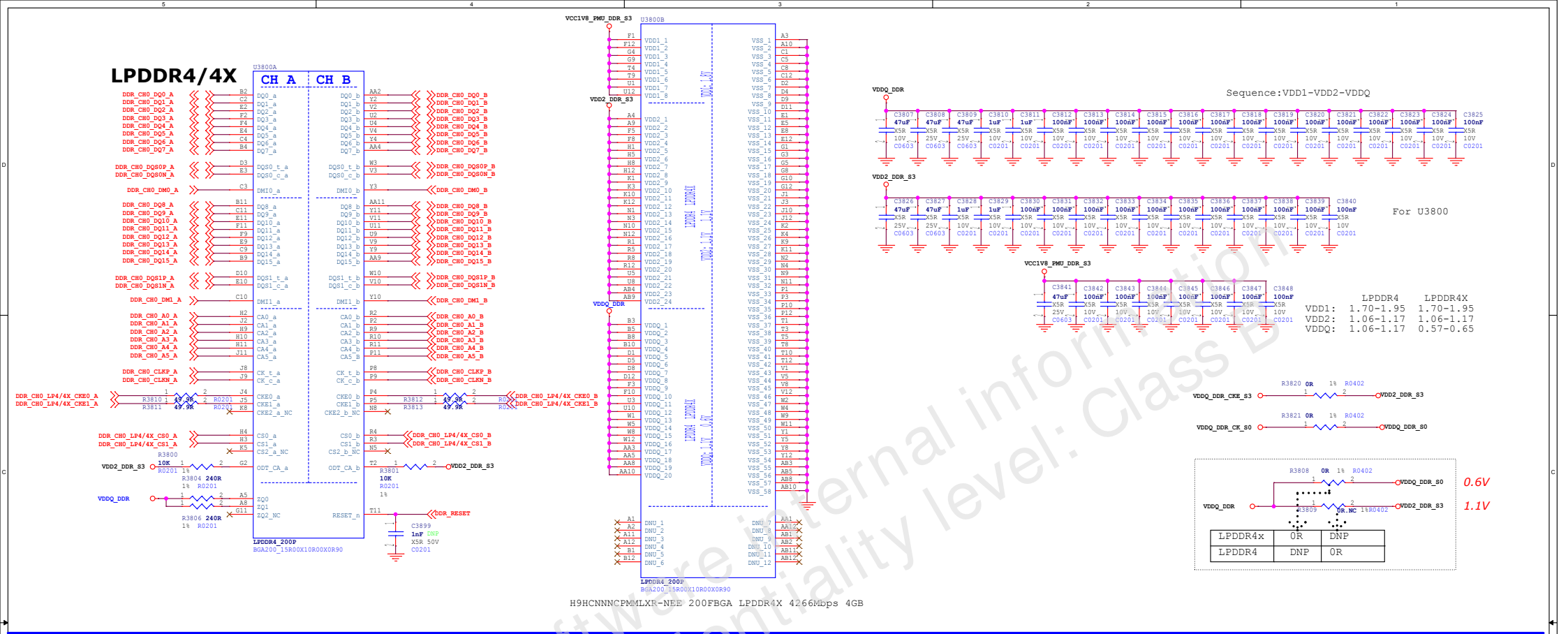


USB3.0 HOST

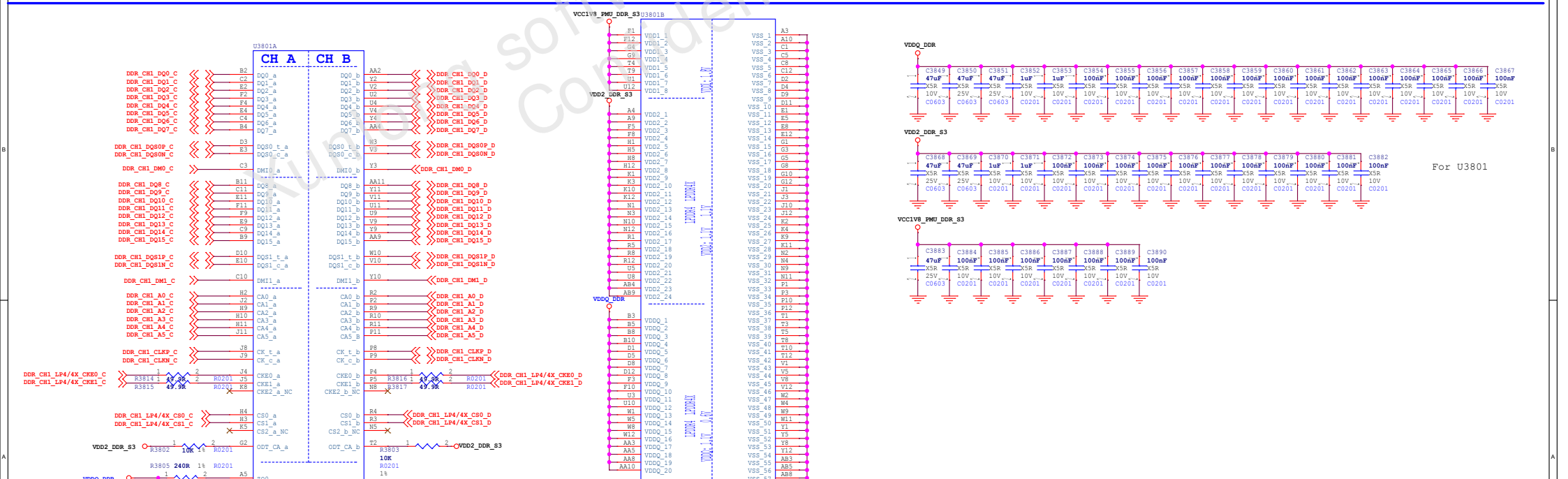


Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: USB20/USB30 HOST Port	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 23 of 36	

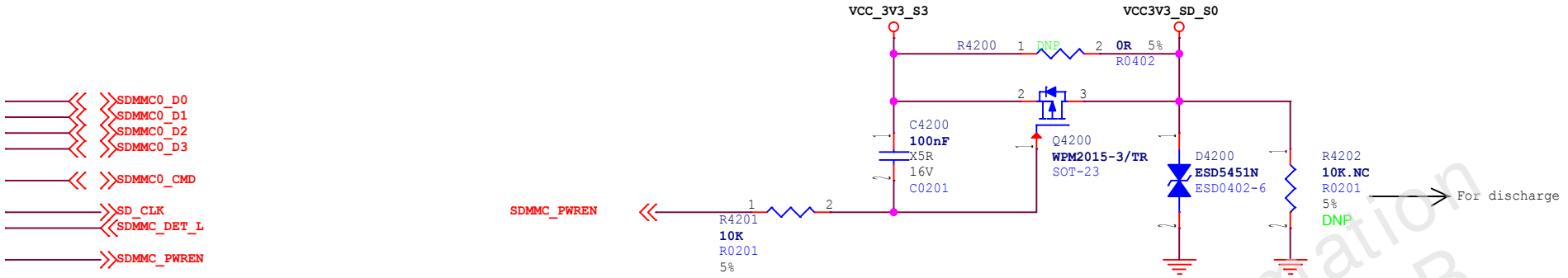


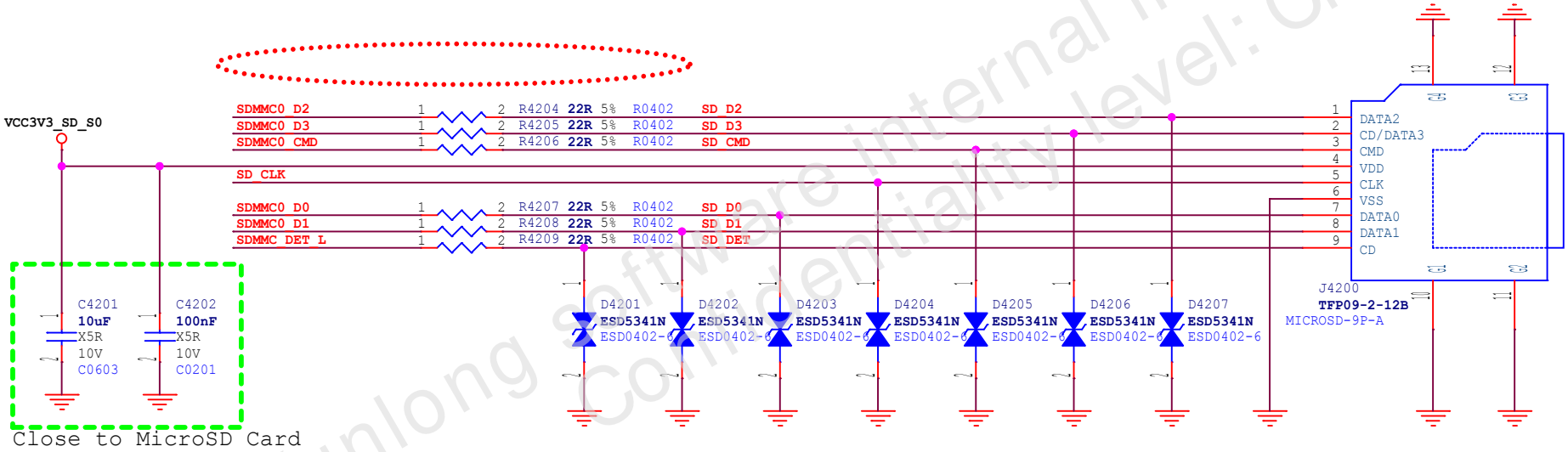
H9HCNNNCPMMLXR-NEE 200FBGA LPDDR4X 4266Mbps 4GB



Nuvoton software internal information
Confidentiality level: Class B

TF CARD





Xunlong software internal information
Confidentiality level: Class B

MicroSD Card

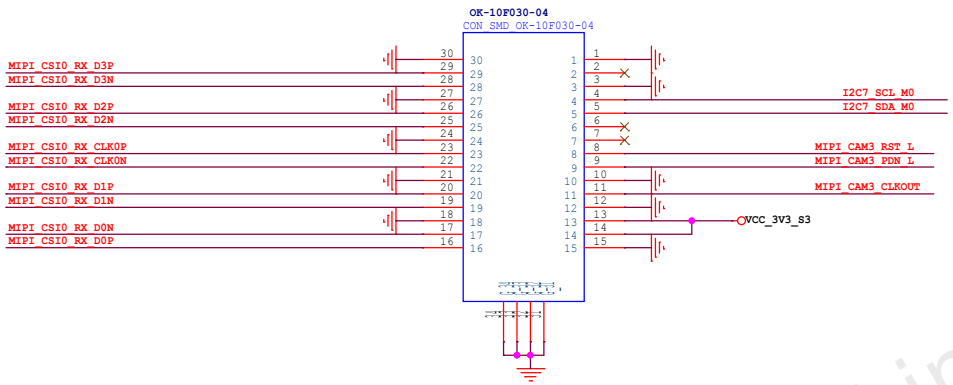
Rockchip Confidential

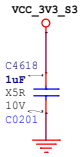
Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A4	Page Name: Flash-TF Card	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 26 of 36	

MIPI-CSI0_RX

MIPI_CSI0_RX_D0P
MIPI_CSI0_RX_D0N
MIPI_CSI0_RX_D1P
MIPI_CSI0_RX_D1N
MIPI_CSI0_RX_D2P
MIPI_CSI0_RX_D2N
MIPI_CSI0_RX_D3P
MIPI_CSI0_RX_D3N
MIPI_CSI0_RX_CLK0P
MIPI_CSI0_RX_CLK0N
MIPI_CAM3_CLKOUT
MIPI_CAM3_RST_L
MIPI_CAM3_PDN_L
I2C7_SCL_M0
I2C7_SDA_M0

Cam1





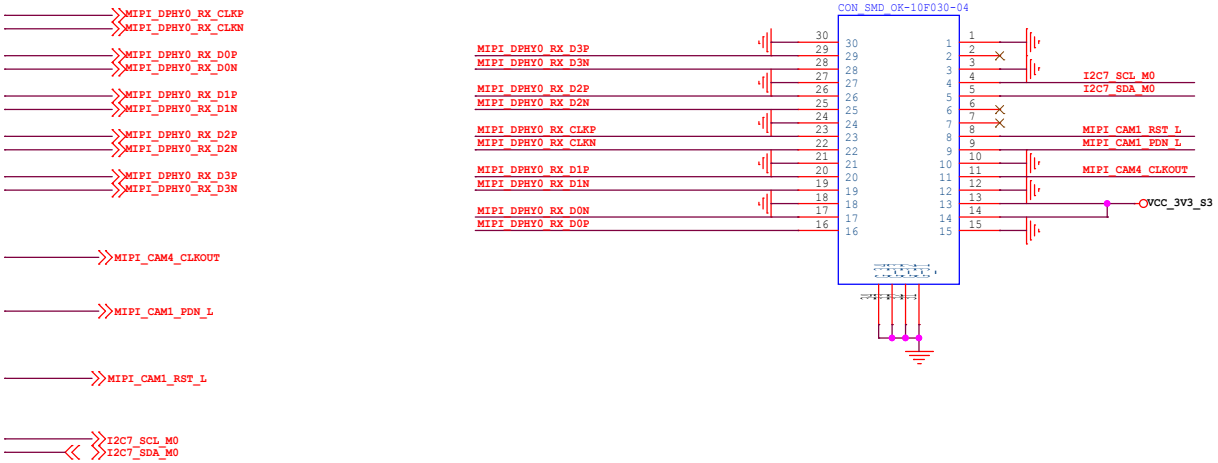
Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: VI-Camera MIPI CSI0-RX / Debug	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 27 of 36	

VI-Camera DPHY_RX

MIPI D-PHY0

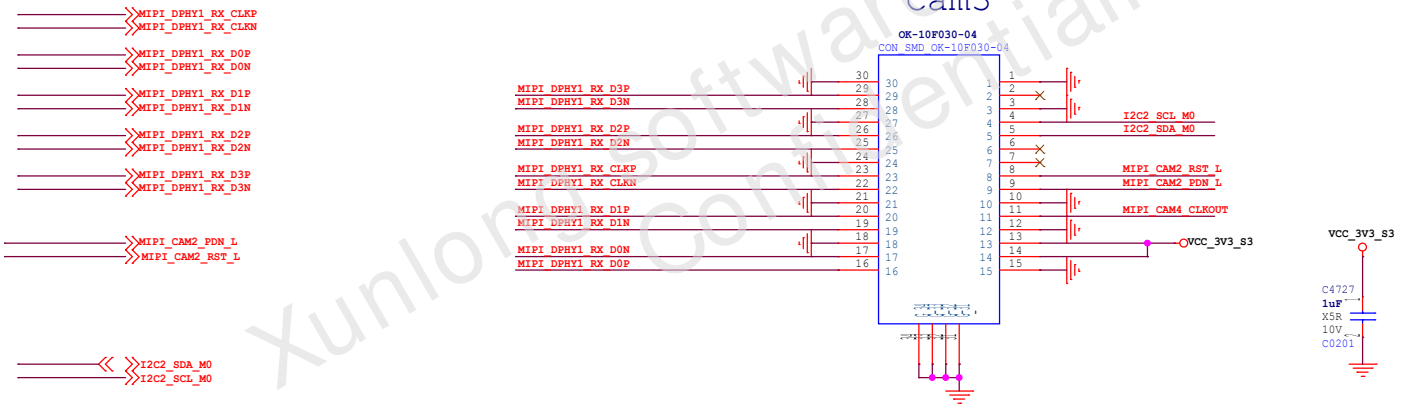
Cam2



VI-Camera DPHY_RX

MIPI D-PHY1

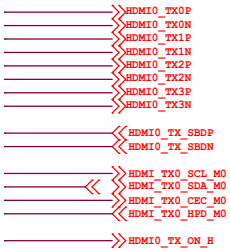
Cam3

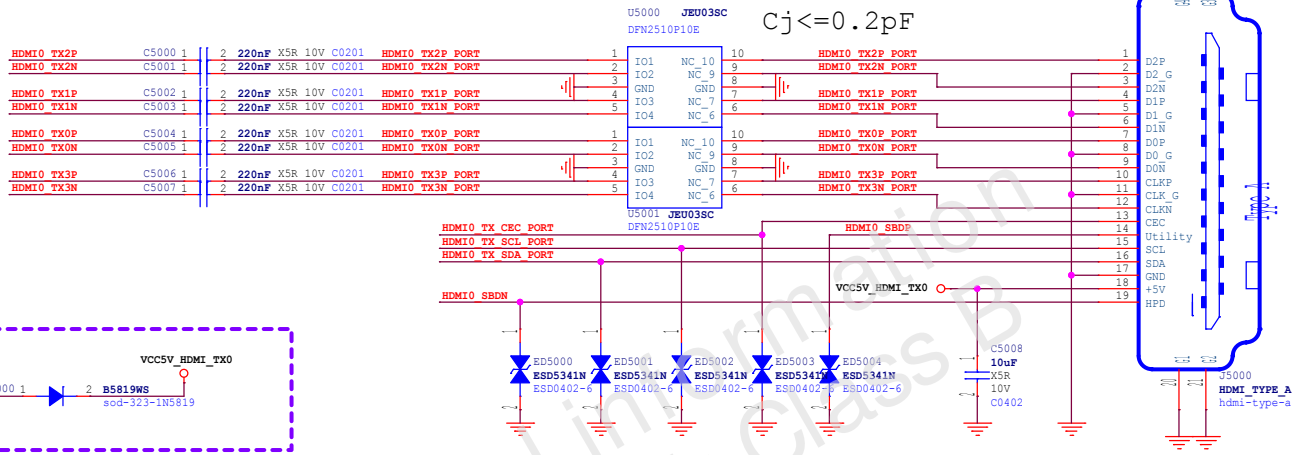


Xunlong software internal information
Confidentiality level: Class B

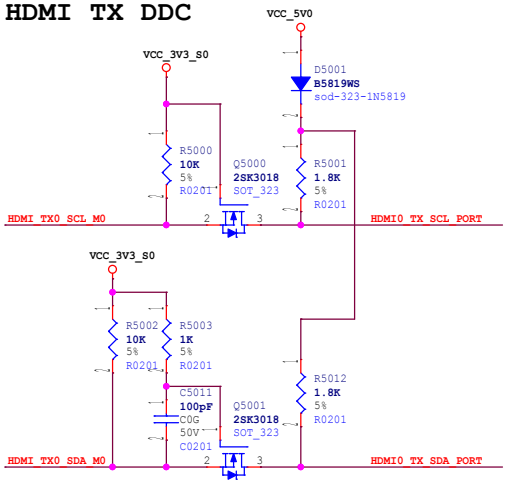
Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: VI-Camera_MIPI-DPHY-RX	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 28 of 36	

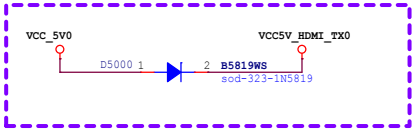
HDMI2.1 TX0



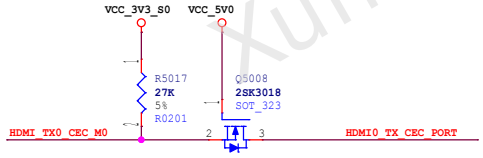


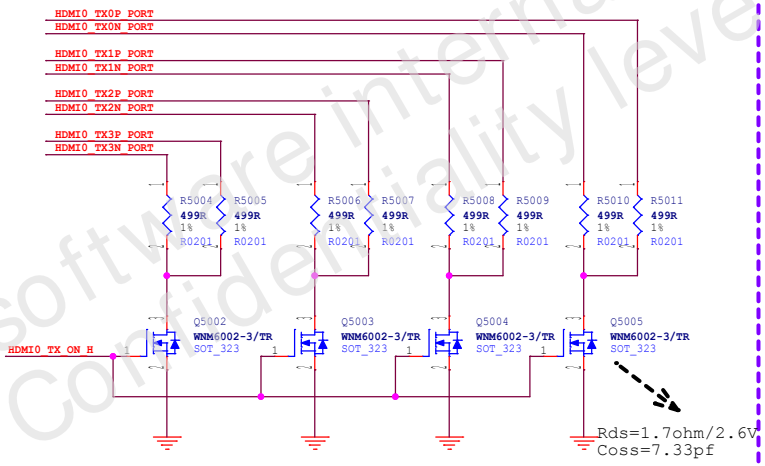
HDMI TX DDC





HDMI TX CEC

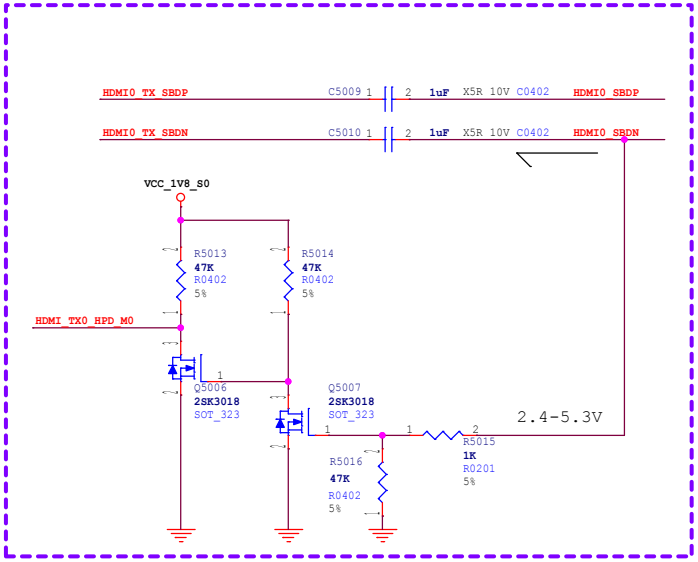




Note:

The controller only support AC coupled link In order to backward compatibility or to meet HDMI2.0 (1.4b) DC common mode spec and Voff, need do R based level-shift.

Switch on in HDMI2.0(TMDS) mode
Switch off in HDMI2.1(FRL) mode.



Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: VO-HDMI2.1 TX	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 29 of 36	

MIPI DPHY TX

SARADC_VIN4_LCD_ID
SARADC_VIN2_LCD_ID

TP_RST_L
TP1_RST_L

LCD_MIPI_PWREN_H
LCD_MIPI_PWREN_H_1

LCD_MIPI_BL_PWM2_M0
LCD_MIPI_BL_PWM6_M0

MIPI_TE0
MIPI_TE1

I2C7_SCL_M0
I2C7_SDA_M0

I2C2_SDA_M0
I2C2_SCL_M0

LCD_MIPI_RESET_0
LCD_MIPI_RESET_1

TP_INT_L
TP1_INT_L

MIPI DPHY0 TX

LCD1

CNN30_1R00_FP05SL_V
FPC30-05PH-BOT

MIPI_DPHY0_TX_D0P
MIPI_DPHY0_TX_D1N
MIPI_DPHY0_TX_CLKP
MIPI_DPHY0_TX_D2N
MIPI_DPHY0_TX_D3P
LCD_MIPI_BL_PWM2_M0
VCC3V3_LCD_MIPI
SARADC_VIN2_LCD_ID
I2C7_SCL_M0
TP_INT_L

MIPI_DPHY0_TX_D0N
MIPI_DPHY0_TX_D1P
MIPI_DPHY0_TX_CLKN
MIPI_DPHY0_TX_D2P
MIPI_DPHY0_TX_D3N
MIPI_TE0
LCD_MIPI_RESET_0
LCD_MIPI_PWREN_H
I2C7_SDA_M0
TP_RST_L

VCC3V3_LCD_MIPI
C5500 10uF X5R 10V C0603
C5501 100nF X5R 16V C0201

VCC5V0_SYS
VCC_5V0

VCC5V0_SYS
C5502 100nF X5R 16V C0201
C5503 22uF X5R 10V C0603

FPC Pin List

Pin1 :GND
Pin2 :D0N
Pin3 :D0P
Pin4 :GND
Pin5 :D1N
Pin6 :D1P
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0

MIPI DPHY1 TX

LCD2

CNN30_1R00_FP05SL_V
FPC30-05PH-BOT

MIPI_DPHY1_TX_D0P
MIPI_DPHY1_TX_D1N
MIPI_DPHY1_TX_CLKP
MIPI_DPHY1_TX_D2N
MIPI_DPHY1_TX_D3P
LCD_MIPI_BL_PWM6_M0
VCC3V3_LCD_MIPI_1
SARADC_VIN4_LCD_ID
I2C2_SCL_M0
TP1_INT_L

MIPI_DPHY1_TX_D0N
MIPI_DPHY1_TX_D1P
MIPI_DPHY1_TX_CLKN
MIPI_DPHY1_TX_D2P
MIPI_DPHY1_TX_D3N
MIPI_TE1
LCD_MIPI_RESET_1
LCD_MIPI_PWREN_H_1
I2C2_SDA_M0
TP1_RST_L

VCC3V3_LCD_MIPI
C5507 10uF X5R 10V C0603
C5508 100nF X5R 16V C0201

VCC5V0_SYS

VCC5V0_SYS
C5509 100nF X5R 16V C0201
C5510 22uF X5R 10V C0603

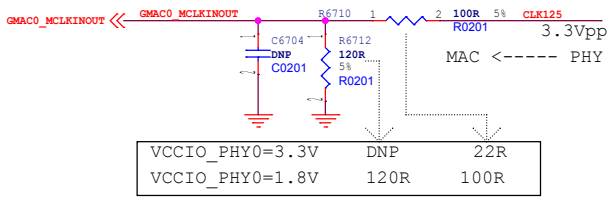
FPC Pin List

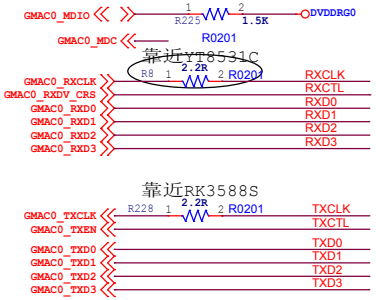
Pin1 :GND
Pin2 :D0N
Pin3 :D0P
Pin4 :GND
Pin5 :D1N
Pin6 :D1P
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0

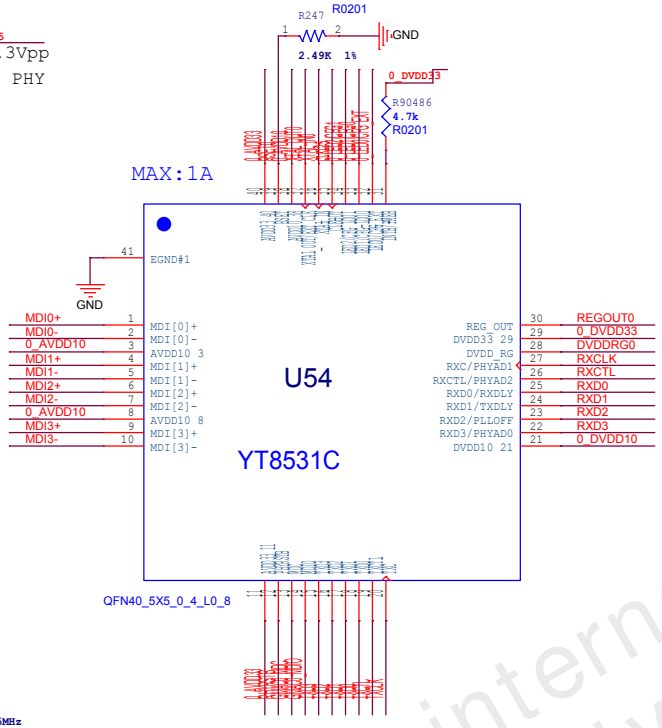
Xunlong software internal information
Confidentiality level: Class B

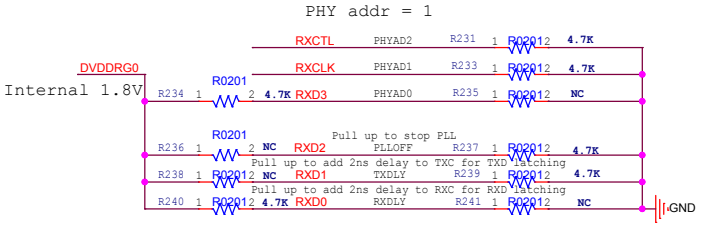
Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: VO-MIPI DPHY0/1-TX	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 30 of 36	

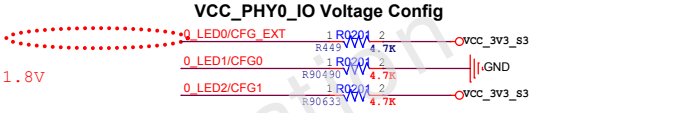
10M/100M/1000M ETH



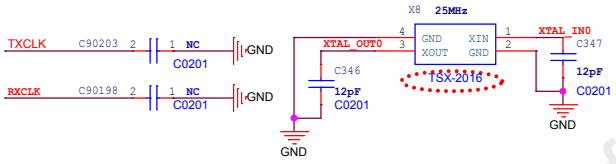


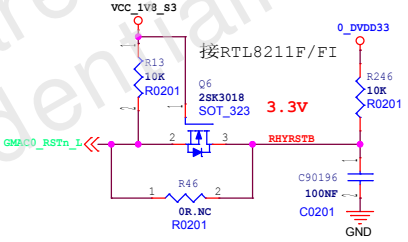


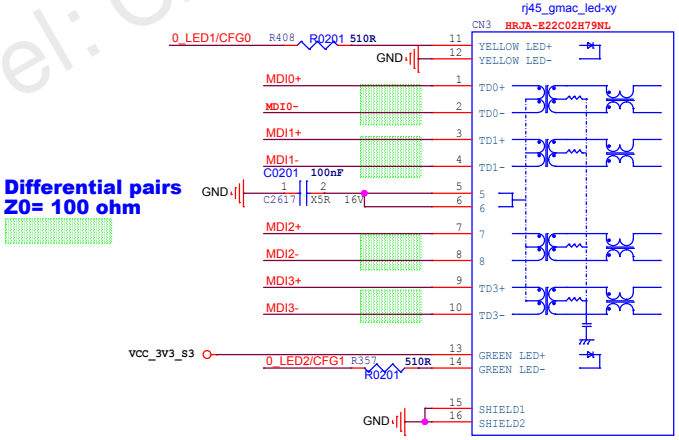




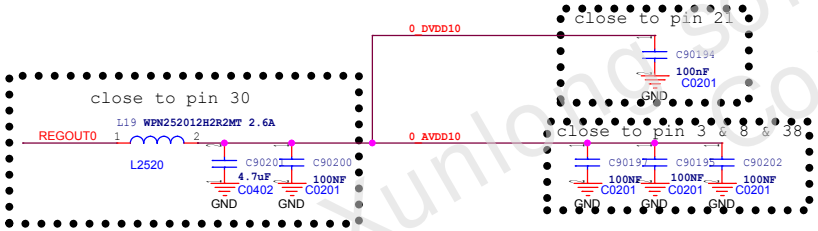
RGMII Power Source	CFG_EXT	CFG_LDO[1:0]	
External 3.3V	1'b1	2'b00	CFG_EXT: 1:External Power Source for IO pad. 0:Integrated LDO for IO pad
External 1.8V (default)	1'b1	2'b10	CFG_LDO(1:0) 10:1.8V 00:3.3V
Internal 1.8V	1'b0	2'b10	

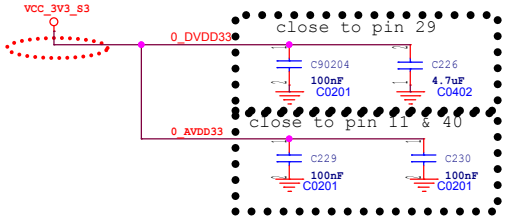


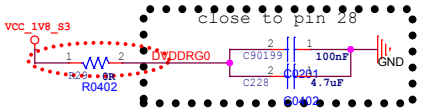


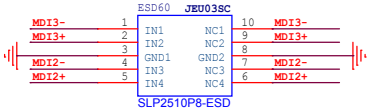


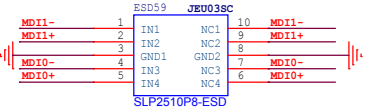
Differential pairs
Z0= 100 ohm







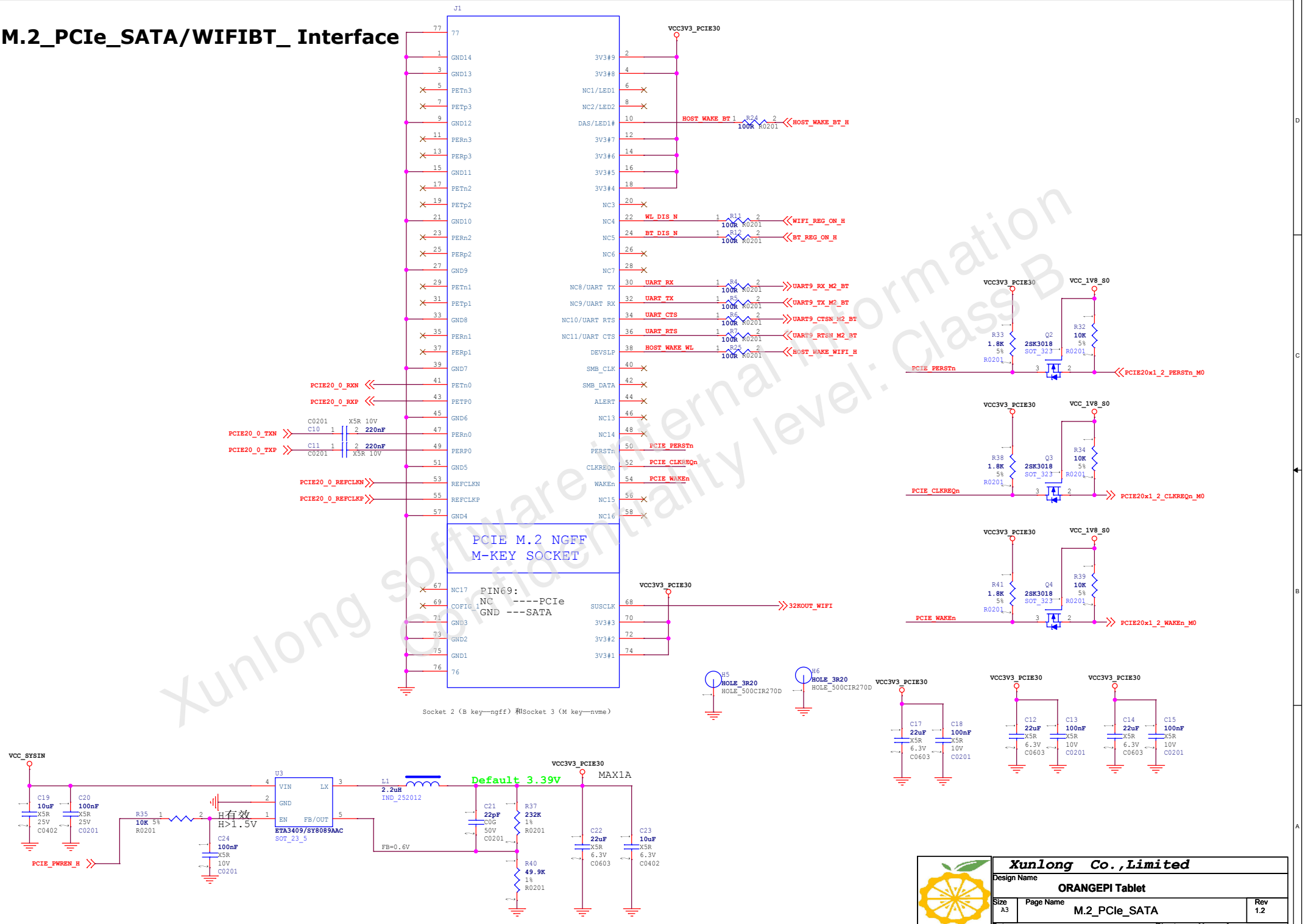




Xunlong software internal information
confidentiality level: Class B

Xunlong Co.,Limited		
Design Name: ORANGEPI 5		
Size: A3	Page Name: 1000M PHY ETH	Rev: 1.2
Date: Tuesday, November 08, 2022	Sheet 31 of 36	

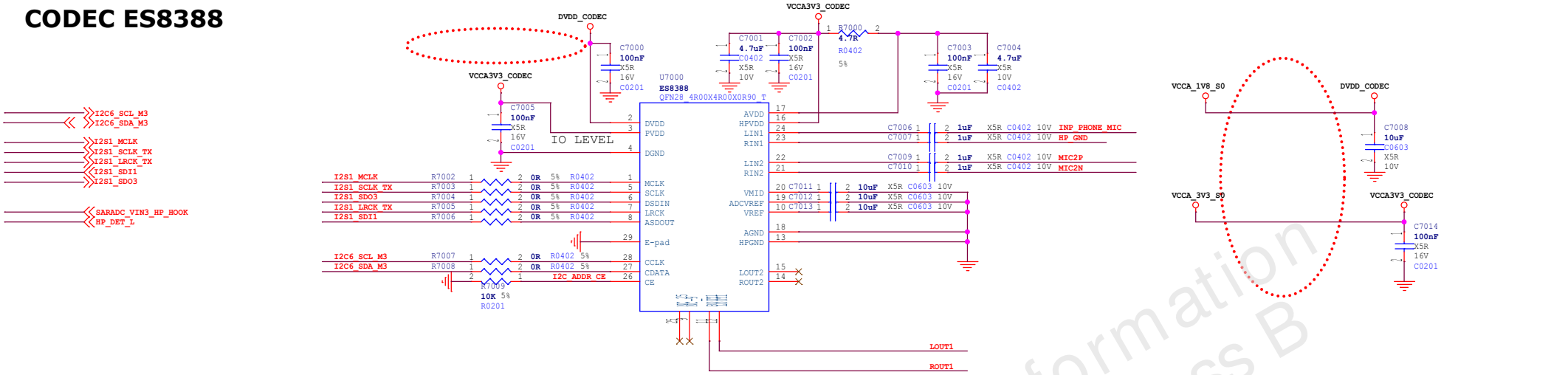
M.2_PCIe_SATA/WIFIBT_ Interface



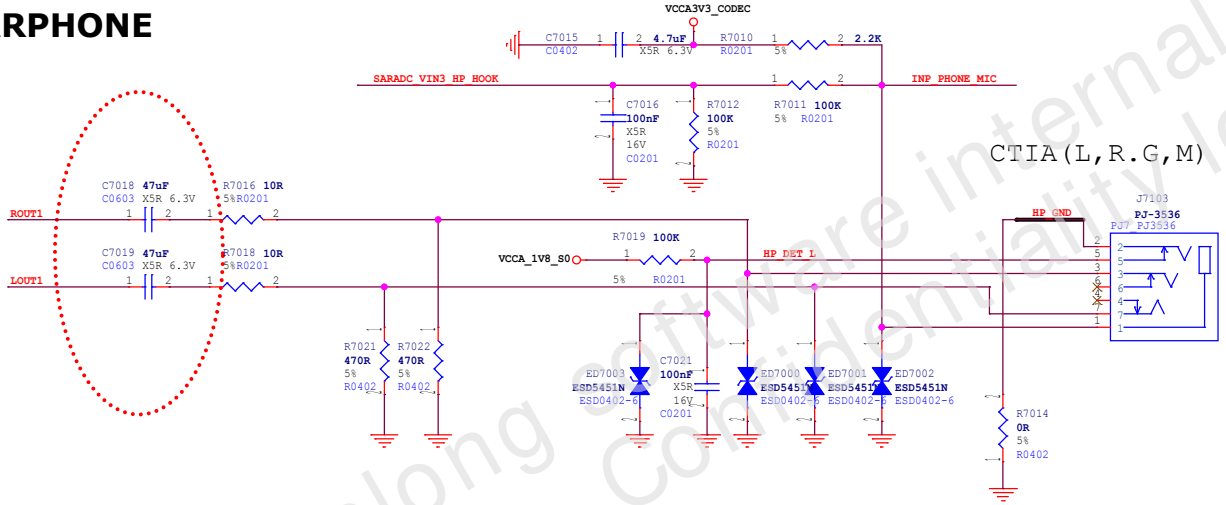
Xunlong software internal information
Confidentiality level: Class B

Xunlong Co.,Limited			
Design Name	ORANGEPI Tablet		
Size A3	Page Name M.2_PCIe_SATA		Rev 1.2
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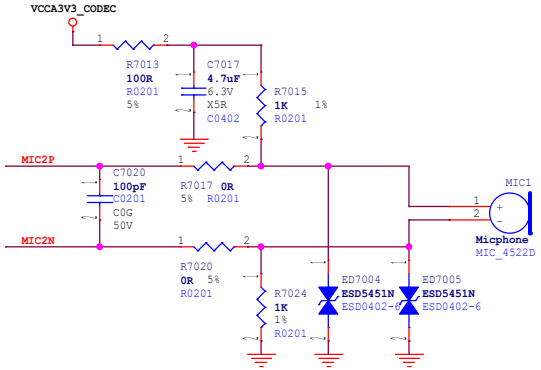
CODEC ES8388



EARPHONE



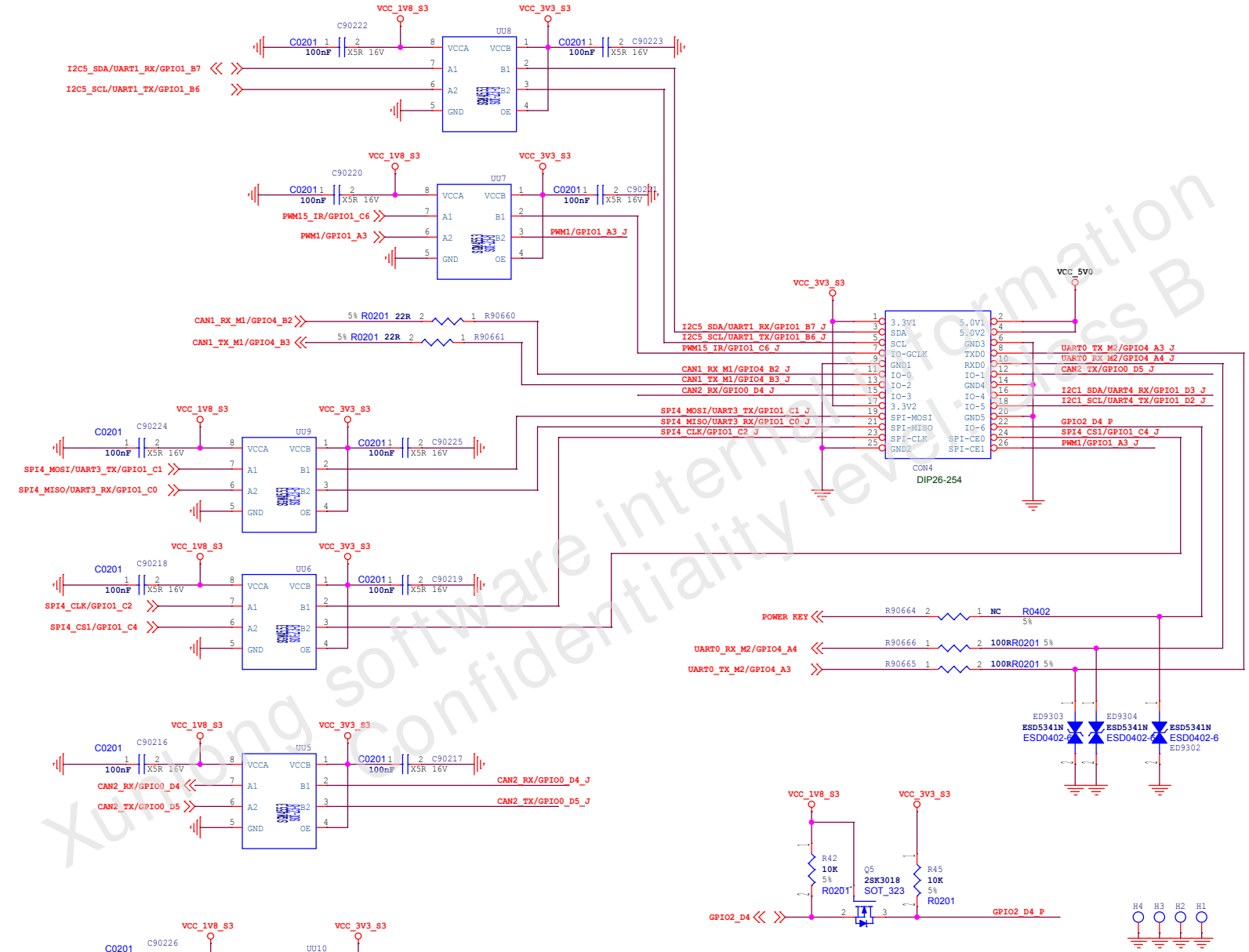
Analog MIC



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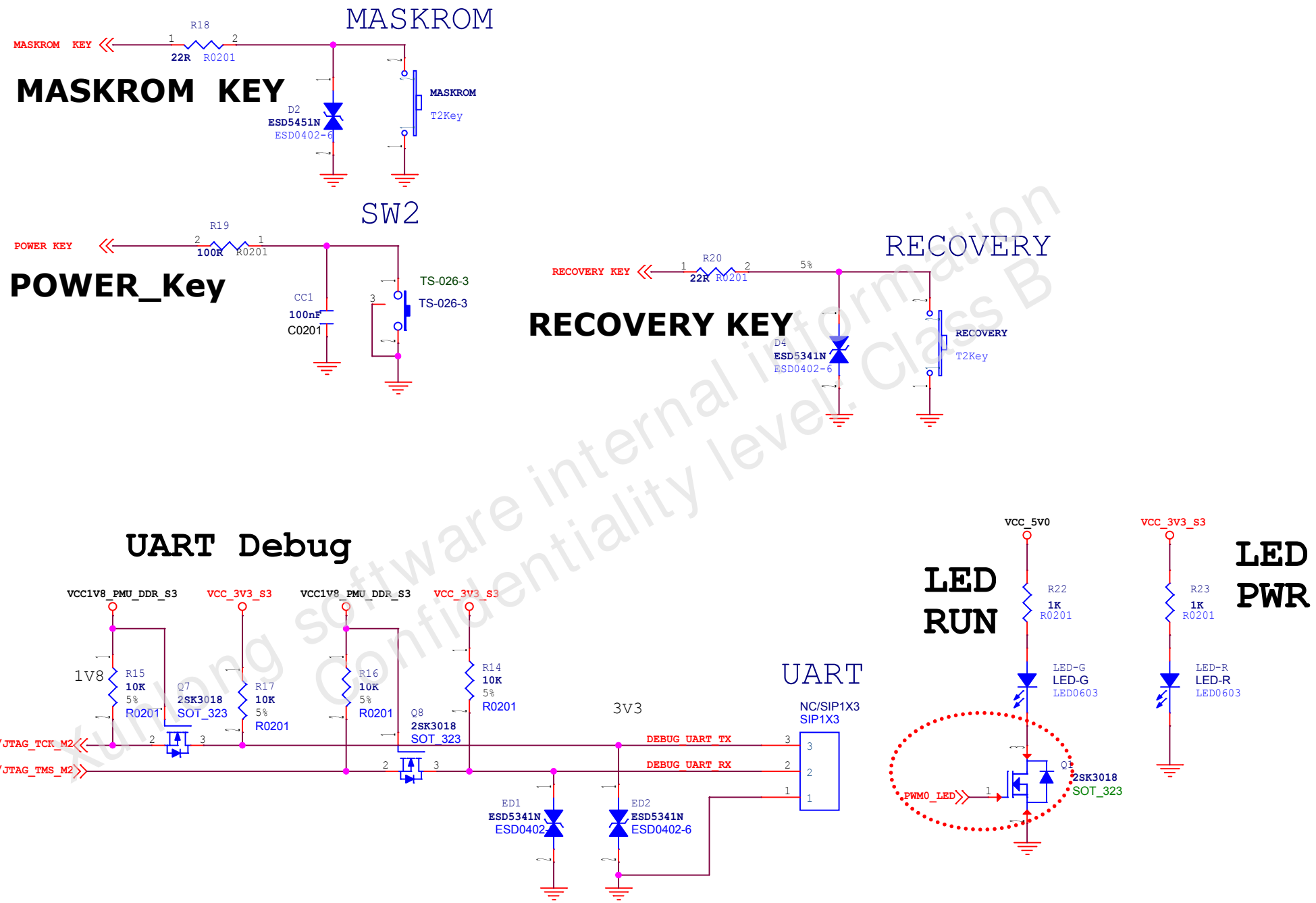
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Design Name	ORANGEPI 5	
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EXT I/O



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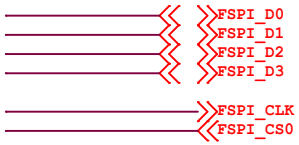
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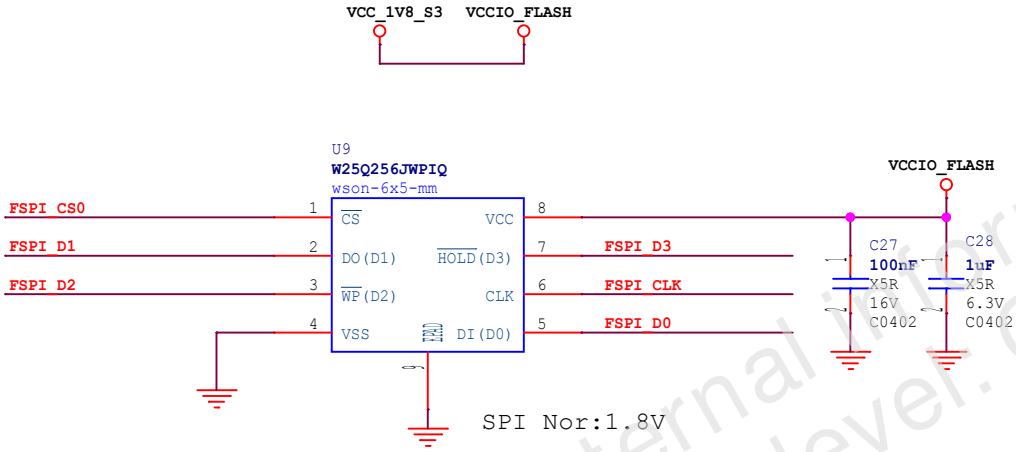


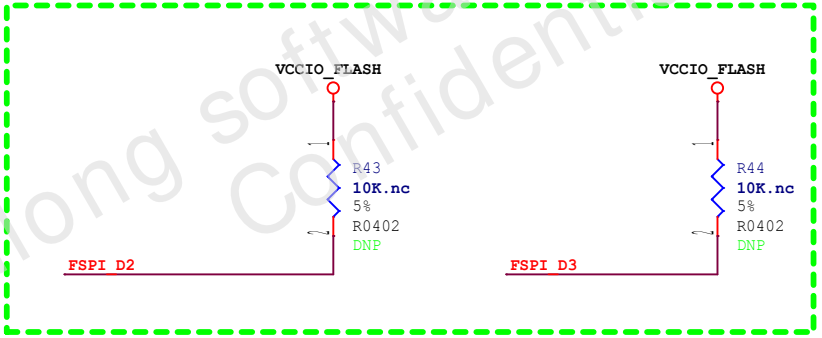
Xunlong software internal information
Confidentiality level: Class B

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SPI FLASH







Note:
When using SPI FLASH with only 1 bit, it needs to be stuffed.

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